
**Cortex®-M4 32-bit Microcontroller with FPU, 200 MHz,
Up to 2048 KB Dual Bank Code Flash, 32 KB Data Flash,
256 KB SRAM, 3-Phase PWM, 12-bit ADC for Motor
Control Application**

DS Rev. 1.00

Features

Core and Memory

- Arm® 32-bit Cortex®-M4F Core
- Up to 200 MHz with FPU, 258 DMIPS
- Cache buffer (flash accelerator) support

Memories

- Up to 2048 KB Code Flash Memory
- Dual Bank Code Flash (bank0, bank1) for OTA (Over-The-Air)
- Bank swap with RWW (Read-While-Write) in Code Flash Memory
- 32 KB of data flash
- Up to 256 KB SRAM
- ECC (Error Correct Code)

Reset and Power Management

- Power-on/power-down reset (POR/LVR/LVI)
- Low-power modes: SLEEP, DEEP-SLEEP

Clock Management

- High-Speed Internal (HSI) oscillator 16 MHz
 - $\pm 3.0\%$ @ -40°C to $+85^{\circ}\text{C}$
 - $\pm 5.0\%$ @ -40°C to $+105^{\circ}\text{C}$
- Low-Speed Internal (LSI) oscillator 500 kHz
 - $\pm 30\%$ @ -40°C to $+105^{\circ}\text{C}$
- High-Speed External (HSE) main oscillator
 - 4 MHz to 16 MHz
- Low-Speed External (LSE) sub-oscillator
 - 32.768 kHz
- Phase-Locked Loop (PLL) frequency generator
 - Up to 200 MHz

General-Purpose Input/Output (GPIO)

- Up to 105 I/O pins with interrupt capability
- All mappable on external interrupt vectors
- Multi-function pins have up to six selections of functions including GPIO

DMA Controller

- 16-ch
- 8-/ 16-/ 32-bit data transfers
- Compatible with 57 different types of peripherals

12-Bit ADC

- Three independent ADC blocks
- 24 analog input channels
- Adjustable sample time and hold time
- Multiple operating modes:
 - Single conversion
 - Sequence conversion
 - Burst conversion
 - Multiple conversion

Comparator (CMP)

- Equipped with Comparator 0 and 1, each of which has three input sources
- Equipped with Comparator 2 and 3, each of which has one input source
- Signals from the DAC can be input to the comparator.

Internal LDO Regulator

12-Bit DAC

- Two independent DAC blocks
- Two analog output channels
- Comparator reference

16-Bit Timer

- 10 channels
 - 10 input capture channels
 - 10 output channels
- Timer operating modes:
 - Periodic timer mode
 - One-shot mode
 - PWM mode
 - Capture mode
- 10-bit prescaler

Pulse Width Modulation (PWM)

- Ten PWM generators
- PWM signal with 16-bit independent counter

Watchdog Timer (WDT)

- 32-bit down-count timer
- Reset and periodic interrupts
- Eight different prescalers

Free-Run Timer (FRT)

- Two 32-bit free-run timers

Motor Pulse-Width Modulation (MPWM)

- Two MPWM generators
- Generating different waveforms through six channels with high- and low-signals of U, V, and W phases
- 16-bit up-/down-counters
- Eight ADC trigger sources
 - 6-Variable Triggers
 - 2-Fixed Trigger (Bottom, Period Match)

Serial Interfaces

- UART
 - Thirteen 16450 asynchronous serial communication ports
- SPI
 - Twelve synchronous serial communication ports
- I2C
 - Four communication ports
- CANFD
 - Two communication ports
- QSPI
 - One communication port

Random Number Generator (RNG)

- Generation of 32-bit random-number
- Programmable RNG seed

AES-128

- One AES channel
- Compatible with direct memory access (DMA)
- FIFO configurable input/output
- Input/output inversion support

CRC Calculation Unit

- CRC operating modes:
 - CRC-32 (0x04C1_1DB7)
 - CRC-16-USB (0x8005)
 - CRC-16-CCITT (0x1021)
 - CRC-8-CCITT (0x07)
- Support for byte reversal of input/output data
- Compatible with DMA

Development Support

- Serial Wire Debug (SWD), JTAG interfaces

Package

- 120-LQFP-1616 (0.5 mm pitch)
- 100-LQFP-1414 (0.5 mm pitch)
- 64-LQFP-1010 (0.5 mm pitch)

Operating Voltage

- 2.7 V to 5.5 V

Operating Temperature

- Industrial grade (−40°C to +105°C)

Applications

- Inverter motor controller
- Electric Pumps and Fans
- Home Appliances (HA) display controller

Product Selection Table

Table 1. Device Summary

Device Name	Code Flash	Data Flash	SRAM	SPI	UART	I2C	MPWM	ADC	I/O Ports	Package
A34G43AYL2N	2048 KB	32 KB	256 KB	12	13	4	2	24	105	120-LQFP
A34G43AVL2N*	2048 KB	32 KB	256 KB	9	10	4	2	24	85	100-LQFP
A34G43ARL2N*	2048 KB	32 KB	256 KB	4	5	4	2	16	49	64-LQFP
A34G430YL2N*	1024 KB	32 KB	256 KB	12	13	4	2	24	105	120-LQFP
A34G430VL2N*	1024 KB	32 KB	256 KB	9	10	4	2	24	85	100-LQFP
A34G430RL2N*	1024 KB	32 KB	256 KB	4	5	4	2	16	49	64-LQFP

* For available options or further information on the devices with “*” marks, please contact the [ABOV sales office](#).

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1. Descriptions

A34G43x series is a 32-bit microcontroller based on the high-performance Arm Cortex-M4F core, with up to 2048 KB of dual bank code flash memory (bank0: 1024 KB, bank1: 1024 KB), 32 KB of data flash memory, and 256 KB of SRAM. It is a powerful microcontroller with a Floating-Point Unit (FPU) that provides effective solutions for various electrical appliances requiring low power consumption and high performance, such as refrigerators, washing machines, dryers, dishwashers, water purifiers, and blenders. It is also suitable for controlling inverter motors where high-performance is critical.

The A34G43x series also offers bank swap with RWW (Read-While-Write), which allows for simultaneous programming while the system continues to operate. Used in conjunction with dual-bank flash memory, the A34G43x series allows code to be executed on one bank while being erased or programmed on the other bank. These functions are effective for implementing features such as OTA (Over-The-Air programming).

1.1 Product Category Definition

Table 2 provides an overview of the memory capacity for available devices in the A34G43x series.

This document provides an overview of the features supported by each device, including high-level information and brief explanations for each feature. Refer to Table 3 for the list of features supported by each product category.

Table 2. A34G43x Series Memory Density

Memory Density		Category
Flash	RAM	
2048 KB	256 KB	A34G43A
1024 KB	256 KB	A34G430

1.2 Availability of Peripherals

Table 3 summarizes and lists product specific features available in the A34G43x series considering the largest package.

Table 3. A34G43x Series Features and Peripherals

Peripherals		Description
Core	CPU	• Maximum operating frequency: 200 MHz • 32-bit Arm Cortex-M4F core - 32-bit Thumb[®]-2 instruction set • Register settings in CPU: - General-purpose registers specified - Main stack pointer (MSP) and process stack pointer (PSP): R13 - Link register (LR): R14 - Program counter (PC): R15 • Data ordering format: Little-Endian • Harvard Architecture • AHB / APB
	Interrupt	• NVIC (Nested-Vectored Interrupt Controller) • Up to 84 peripheral interrupts supported • Assignable with 16 different priority levels
	FPU	• Rendered by extending and transforming the ARMv7 floating-point arithmetic functionality • Compliant with the ANSI / IEEE 754 standard • Capable of binary floating-point arithmetic and computation

Table 3. A34G43x Series Features and Peripherals (continued)

Peripherals		Description
Memory	Code Flash	- Capacity: - A34G43A: 2048 KB code flash memory - A34G430: 1024 KB code flash memory - A high-capacity code flash memory built in - Max 33 MHz flash access speed - Two-word (8-byte) program - Bulk (full-chip), 1 KB and 4 KB erases - Read protection - Self-programming (Supports updating data in a code flash memory region during the execution of user program in the code area.) - CRC code generation and verification for the flash memory - Dual Bank code flash memory (bank0: 1024 KB, bank1: 1024 KB) for OTA (Over-The-Air programming) - Support RWW (Read-While-Write) - Support ECC (Error Correct Code) - Endurance: 10,000 cycles - Lifetime: 10 years
	Data Flash	- Capacity: 32 KB - Max 33 MHz access speed - One-word (4-byte) program 512 bytes and 2 KB erases - CRC code generation and verification for the flash memory - Support ECC (Error Correct Code) - Endurance: 100,000 cycles - Lifetime: 10 years
	BootROM	- Entering the boot mode by setting the nBOOT pin to low-level during a reset procedure. - UART boot mode - In-System Programming - The boot mode can program the internal flash memory via UART.
	SRAM	- Capacity: 256 KB - Usable as a program's work area - High-speed execution enables the execution of time-critical codes. - Part of the SRAM can be remapped into an interrupt vector area.

Table 3. A34G43x Series Features and Peripherals (continued)

Peripherals		Description
System Control Unit (SCU)	Clock	• High-speed internal oscillator (HSI) • 16 MHz - $\pm 3.0\%$ @ -40°C to $+85^{\circ}\text{C}$ - $\pm 5.0\%$ @ -40°C to $+105^{\circ}\text{C}$ • Low speed internal oscillator (LSI) - 500 kHz ($\pm 30\%$ at -40°C to $+105^{\circ}\text{C}$) • External main oscillator (HSE): 4 MHz to 16 MHz • External sub-oscillator (LSE): 32.768 kHz • Phase-Locked Loop (PLL) frequency generator generates a high-speed clock (up to 200 MHz)
	Clock Monitoring	• System Fail-Safe function by Clock Monitoring - External main oscillator (HSE) - External sub oscillator (LSE) - Main system clock (MCLK)
	Operating Mode	• Run mode • Sleep mode • Deep sleep (STOP1, STOP2) mode
	Reset	• nRESET pin reset • Core reset • Software reset • POR (Power-On Reset) • LVR (Low Voltage Reset) • WDTR (Watchdog Timer Reset) • Reset due to clock oscillating error • CFMC bank reset
	LDO	• Low-dropout (LDO) regulator built in for low-voltage operation
	POR	• The POR generator detects an internal 1.8 V voltage and generates a reset signal.
	LVI	• 16 low-voltage detection levels • Supports LVD interrupt • Supports wake-up from SLEEP mode

Table 3. A34G43x Series Features and Peripherals (continued)

Peripherals		Description
System Control Unit (SCU)	Wake-up	• Wake-up by a general-purpose input/output (GPIO) pin • Wake-up by a free-run timer (FRT) • Wake-up by a Watchdog Timer (WDT) • Wake-up by a low-voltage indicator (LVI)
General-Purpose I/O (GPIO)		• Input/output (I/O) port for general purposes • 120-LQFP - I/O pins: 105 • 100-LQFP - I/O pins: 85 • 64-LQFP - I/O pins: 49 • Each pin can be set for one of the following modes: - Push-pull output - Open drain output - Input • The use of each pin can be set by setting the pin multiplexer. • Each pin can be configured as an external interrupt source, either the high-level / low-level interrupt or the rising-edge / falling-edge interrupt • Pull-up or pull-down resistor, and debouncing can be set for each pin. • Drive strength can be adjusted for each port pin. • Each pin bit can be individually set and reset. • Wake-up events triggered by external asynchronous inputs
Direct Memory Access Controller (DMA)		• 16-ch direct memory access (DMA) support peripherals • 8- / 16- / 32-bit data transfers • Compatible with 57 different types of peripherals - SPIn (SPI0, SPI1, SPI2, SPI3, SPI4, SPI5, SPI6, SPI7, SPI8, SPI9, SPI10, SPI11) - UARTn (UART0, UART1, UART2, UART3, UART4, UART5, UART6, UART7, UART8, UART9, UART10, UART11, UART12) - CRC - AES-128 - ADC (ADC0, ADC1, ADC2)

Table 3. A34G43x Series Features and Peripherals (continued)

Peripherals		Description
TIMER	16-bit timer	• General-purpose 16-bit up-count timer • 10 channels - 10 timer <i>n</i> capture port (TnIO) input channels - 10 timer <i>n</i> output port (TnIO) output channels • Timer operating modes - Periodic timer mode - One-shot mode - PWM mode - Capture mode • Interrupt events - Timer / counter match interrupt - Timer overflow interrupt • Input clock selection - Clocks are freely selectable through the miscellaneous clock control registers (SCU_MCCR_x) - External clocks are selectable • Timer signals can be generated through TnIO pins • 10-bit prescaler
	WDT	• 32-bit down-count timer • Reset and periodic interrupts • Clocks are freely selectable through the miscellaneous clock control registers (SCU_MCCR_x) • Eight different prescalers are selectable
	FRT	• Two 32-bit free-run timer - Capable of calculating the internal system time - 32-bit up-count timer • Interrupt events - Period interrupt - Overflow interrupt
Pulse-Width Modulation	PWM	• PWM generator with 10 channels • PWM signal with 16-bit independent counter • Consists of 1-unit per 5 channels. • 8-bit prescaler per 1 unit • Configuration of duty and period of PWM output signal • 1/2, 1/4, 1/8, 1/16 clock divider • 16-bit period and count value set • Built-in channels that outputs inverted PWM signal

Table 3. A34G43x Series Features and Peripherals (continued)

Peripherals		Description
Serial Interface	UART	• A total of thirteen 16450 asynchronous serial communication ports • Configurable standard asynchronous communication bits (start, stop, and parity) • Flexible communication available through programming - 5- to 8-bit data transfers - Even / Odd / Non-parity generation and checking - 1-bit, 1.5-bit, or 2-bit stop bit generation and checking • 8-bit fraction controller and 16-bit baud rate generator
	SPI	• Twelve synchronous serial communication port channels • Master / slave operation • Loop-back mode • Programmable and flexible communication • 8- / 9- /16- /17-bit data transmit and receive - SPI clock speed - Both least significant bit (LSB)-first and most significant bit (MSB)-first modes available
	I2C	• Standard I2C communication protocol • Four channels supported • Master and slave modes supported for each channel • 7-bit addressing supported for slave mode • SCL signal's high / low periods and SDA signal's hold time settable
	CANFD	• Supports Flexible Data Rate (CANFD) protocol for high-speed communication. • Backward compatible with Classical CAN. • Configurable data payload sizes (up to 64 bytes). • Multiple message buffers for efficient message handling - Acceptance filter buffer: 16 - Transmit buffer: 16 - Receive buffer: 23 • Supports error detection and fault confinement.
	QSPI	• Supports Quad Serial Peripheral Interface for high-speed flash memory communication. • Dual/Quad SPI modes supported for faster data transfers. • DDR (Double Data Rate) and SDR (Single Data Rate) modes supported for flexible communication. • Configurable clock speed and data lines for optimized performance. • Ideal for executing code or reading data directly from external flash memory (XIP support).

Table 3. A34G43x Series Features and Peripherals (continued)

Peripherals		Description
Motor Pulse-Width Modulation	MPWM	• Two MPWM generators • Six channels (high and low signals of phases U, V, and W) generate different waveforms • 16-bit up / down counters • Eight ADC trigger sources - Six variable triggers - Two fixed triggers (bottom, period match) • Interrupt events - Bottom interrupts - Top (period) interrupts • Interval interrupt mode • Falling / rising dead-time applicable • A special operating mode: • Phases U, V, and W are independently controlled. - Different carrier counters running for phases U, V, and W - Different duty and periods configurable for phases U, V, and W • Protection and over-current detection supported
12-bit A/D Converter	ADC	• Three independent ADC blocks • 24 analog input channels • Operating modes: - Single conversion - Sequence conversion - Burst conversion - Multiple conversion - Inject conversion • Up to eight sequential conversions supported • Software triggers supported • Three internal trigger sources (MPWM and timers) supported • Sample time and hold time are adjustable
12-bit D/A Converter	DAC	• Two independent DAC blocks • Comparator reference supported • 12-bit resolution
Comparator	CMP	• Equipped with Comparator 0 and 1, each of which has three input sources • Equipped with Comparator 2 and 3, each of which has one input source • Signals from the DAC can be fed to the comparator
Advanced Encryption Standard	AES-128	• One AES channel • Compatible with direct memory access (DMA) • Input/output FIFO configurable • Input/output inversion supported

Table 3. A34G43x Series Features and Peripherals (continued)

Peripherals		Description
Random Number Generator	RNG	• Random-Number Generator • Interrupt events • Generator ready interrupt • Error interrupt
Cyclic Redundancy Check	CRC	• CRC operating modes: - CRC-32 (0x04C1_1DB7) - CRC-16-USB (0x8005) - CRC-16-CCITT (0x1021) - CRC-8-CCITT (0x07) • Input/output data reversion supported • Compatible with DMA
Operating Voltage		• 2.7 V to 5.5 V
Operating temperature		• Industrial grade (−40°C to +105°C)
Package		• Three types of package options - 120-pin LQFP - 100-pin LQFP - 64-pin LQFP

Table 4. Summary of A34G43x Peripherals

Peripheral		A34G43xYL	A34G43xVL	A34G43xRL
Code Flash Memory		A34G43A: 2048 KB A34G430: 1024 KB		
Data Flash Memory		32 KB	32 KB	32 KB
SRAM		256 KB	256 KB	256 KB
Timers	MPWM	2 (16-bit)		
	General-Purpose	10 (16-bit)		
	PWM ¹⁾	10 (16-bit)		
	SysTick Timer	1		
	Watchdog Timer	1		
	Free-Run Timer	2		
Communication Interfaces	SPI	12	9	4
	I2C	4	4	4
	UART	13	10	5
	CANFD	2	2	2
	QSPI	1	1	1
RNG		1		
DMA (Direct Memory Access)		16		
GPIO		105	85	49
12-bit ADCs		1.5 Msps x 3 units		
Number of Channels		24	24	16
DAC		2		
CMP ²⁾		4		
RNG		1		
AES-128		1		
CRC		1		
Operating Frequency		Max. 200 MHz		
Operating Voltage		2.7 V to 5.5 V		
Operating Temperature		Ambient operating temperature: -40°C to +105°C		
Packages		120-LQFP	100-LQFP	64-LQFP

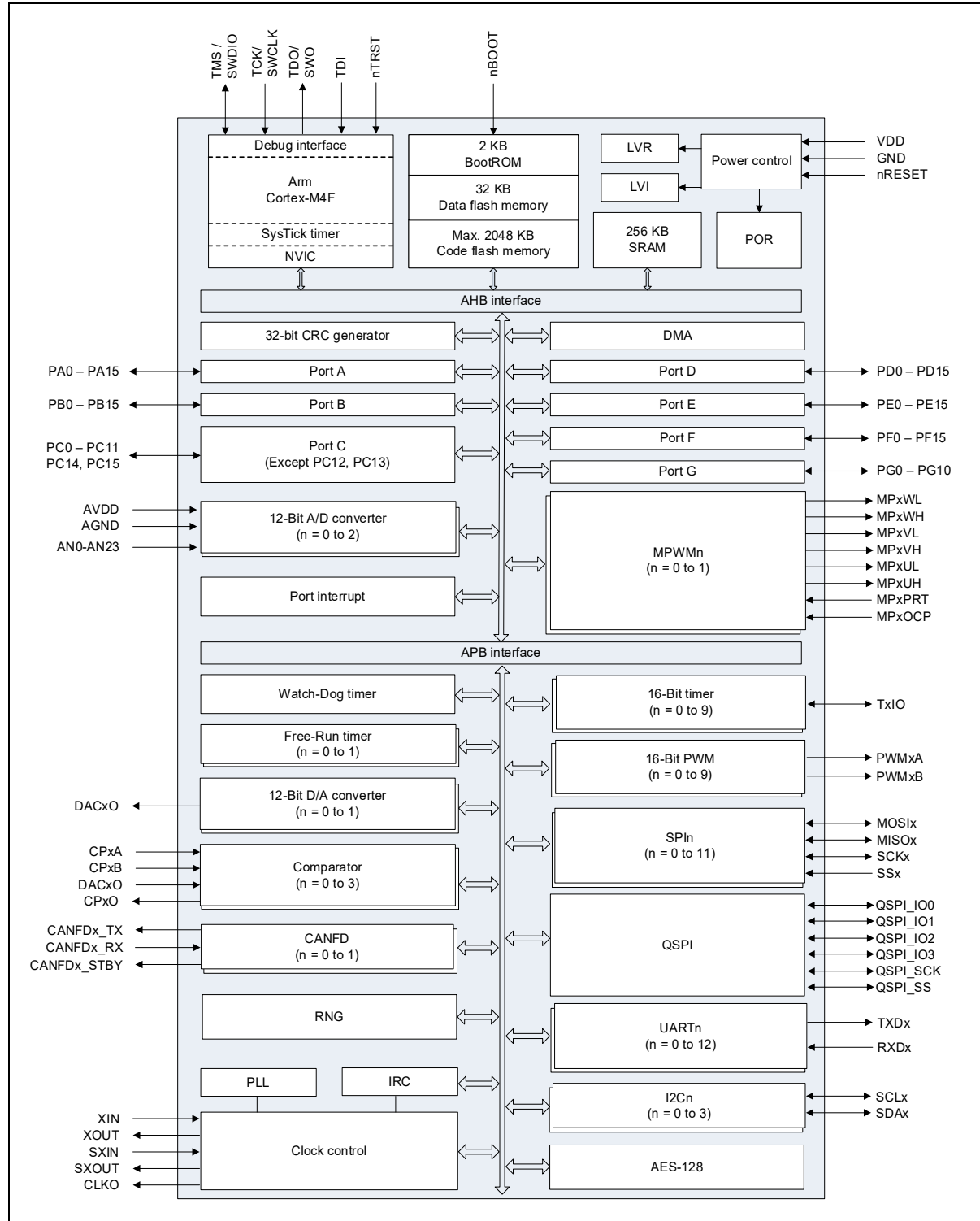
NOTES:

1. In the A34G43xRL (64-LQFP) package, there are no external pins available for PWM6B, PWM7B, and PWM8B.
2. In the A34G43xRL (64-LQFP) package, external pins for CMP2 and CMP3 are not assigned. They can only be used for the internal DAC or V_{REFINT} .

1.3 Block Diagram

Figure 1 shows a block diagram of the A34G43x series.

Figure 1. A34G43x Block Diagram



1.4 Functional Overview

The following sections provide overviews of the features of the A34G43x series microcontroller.

1.4.1 Cortex-M4F Core

The Arm® Cortex-M4F has the same functions as the Cortex-M4 and includes optional floating point arithmetic functionality. The two processors are intended for deeply embedded applications that require fast interrupt response features.

Refer to the technical reference manual “ARM ID061113” for detailed information of Cortex-M4F.

1.4.2 Floating Point Unit (FPU)

Cortex-M4 FPU is an implementation of the single precision variant of the ARMv7-M Floating-Point Extension (FPv4-SP). It provides floating-point computation functionality that is compliant with the ANSI / IEEE Std 754-2008, IEEE Standard for Binary Floating-Point Arithmetic, referred to as the IEEE 754 standard. The FPU supports all single-precision data-processing instructions and data types described in the ARM®v7-M Architecture Reference Manual.

1.4.3 Embedded SRAM

The A34G43x series has a block of zero-wait on-chip SRAM. The size of the SRAM is 256 KB, and its base address is 0x2000_0000.

This SRAM memory area is usually used for data memory and stack memory. Sometimes the code is dumped into the SRAM memory for fast operation or flash erase / programming operation.

1.4.4 Boot Configuration

The A34G43x series has a boot mode option to program internal flash memory. Boot mode can be entered by setting nBOOT pin to 'L' at reset timing (Normal state is 'H').

Boot mode supports UART boot. The pins for the boot mode are listed in Table 5.

Table 5. Boot Mode Pin List

Block	Pin Name	Pin Direction	Description
SYSTEM	nRESET/PC10	Input	Reset input signal
	nBOOT/PC11	Input	Boot mode setting pin
UART0	RXD0/PC14	Input	UART boot receive data
	TXD0/PC15	Output	UART boot transmit data

1.4.5 Power Supply

The device requires a 2.7 V to 5.5 V operating voltage supply (VDD) and analog peripherals are supplied through the independent power domain AVDD.

Table 6. Power Supply

Parameter	Symbol	Min	Max	Unit
Supply Voltage	V _{DD}	2.7	5.5	V
	AV _{DD} (Analog V _{DD})	2.7	5.5	V

During a power-up process, a reset is fundamental and affects the entire system booting process. A34G43x series has two power-related reset options as described below:

- POR_RST (Power-On Reset) that controls the voltage less than 1.8 V.
- LVD_RST (Low-Voltage Detect Reset) that controls the voltage less than 2.11 V (default).

To prevent this abnormal code read operation, the LVD_RST generates the SYSRESETn internal signal and enter the microcontroller into reset mode to prevent the abnormal operation.

1.4.6 Operation Mode

1.4.6.1 Transition of Operation Mode

INIT mode is the initial state of the chip when a reset signal is asserted.

In RUN mode, the CPU can operate at the maximum performance with the high-speed clock system.

SLEEP and DEEP-SLEEP modes can be used as low power consumption mode. In the low power consumption mode, power is effectively managed to reduce power consumption by halting the processor core and any unused peripherals.

1.4.6.2 RUN Mode

In RUN mode, the CPU and peripheral hardware operate with a high-speed clock. If the INIT state occurs after a reset, the system enters RUN mode.

1.4.6.3 SLEEP Mode

Once the microcontroller enters SLEEP mode, the CPU becomes inactive. By setting the SCU_PER and SCU_PCER registers, users can determine which peripherals are to be inactive in SLEEP mode.

1.4.6.4 DEEP-SLEEP (STOP1, STOP2) Mode

When the core goes under stop state using WFI instruction, the chip enters in DEEP-SLEEP mode. DEEP-SLEEP mode is divided into DEEP-SLEEP (STOP 1) and DEEP-SLEEP (STOP 2) modes.

1.4.6.5 Reset Mode

The A34G43x series has two system reset options: a cold reset, which is effective during power up or down sequences, and a warm reset, which is triggered by multiple reset sources.

The reset features of A34G43x series are as follows:

- Power-On reset (POR)
- Low-Voltage Detect Reset (LVR)
- nRESET pin reset
- Watchdog Timer (WDT) Reset
- Software reset
- Clock oscillating error reset
- CPU request reset
- CFMC bank reset

1.4.7 Clocks and Startup

The A34G43x series has two main operating clocks: HCLK, which generates clock signals for CPU and AHB system, and PCLK, which generates clock signals for peripheral systems.

1.4.7.1 HCLK Clock

The Cortex-M4F core requires two clocks: HCLK and FCLK. While the HCLK supplies both the CPU and AHB clocks, the FCLK remains enabled unless the system enters DEEP-SLEEP mode. Conversely, during SLEEP mode, the HCLK can be disabled while the FCLK continues to function.

The buses and memories are clocked by the HCLK. As the bus clock frequency is limited to a maximum of 200 MHz, the HCLK frequency must not exceed 200 MHz.

1.4.7.2 PCLK Clock

The PCLK is used as a clock source for peripherals. The SCU_PCER registers can enable or disable the PCLK for each peripheral. If a peripheral block's PCLK input is not enabled, its registers cannot be read. Additionally, it is important to note that the PCLK stops operating in DEEP-SLEEP mode.

1.4.7.3 Clock Configuration Procedure

When the microcontroller is powered up, the LSI (500 kHz) operates as the default system clock. Later, during system initialization, the system clock changes from LSI (500 kHz) to HSI (4 MHz = 16 MHz/4). HSI (16 MHz) and HSE can be configured under user control using the SCU_CSCR (clock source control register). After enabling the HSE block, it is necessary to wait for the crystal oscillation to be stabilized before proceeding.

The sub-oscillator, Low Speed External (LSE, operating at 32.768 kHz) clock, can be enabled using the SCU_CSCR register (Clock Source Control Register). Like the HSE, the LSE must be enabled and allow sufficient time for the oscillation to be stabilized.

NOTE:

1. XIN, XOUT, SXOUT, and SXIN cannot be used for GPIO or other alternate functions. When the HSE and LSE clocks are not in use, the clocks must be disabled by using software.

1.4.8 Interrupts and Events

1.4.8.1 Nested Vectored-Interrupt Controller (NVIC)

The A34G43x series includes up to 84 maskable interrupt sources that manage sixteen priority levels and the Nested Vector Interrupt Controller (NVIC) that controls the interrupt lines of the Cortex®-M4.

The NVIC has the following advantages:

- NVIC enables interrupt processing while reducing interrupt latency.
- Interrupt Vector Table (IVT) address delivered directly to the core.
- Fast interrupt processing capability
- Higher priority interrupts are completed before lower priority interrupts, although they are generated later than the lower priority interrupts.
- Tail Chaining
- Automatic saving of processor state
- Interrupt restoring without instructional overhead at the end of the interrupt

The NVIC hardware block provides flexible and efficient interrupt management with minimal interrupt latency.

1.4.8.2 Extended Interrupt (EXTI)

The Extended Interrupt feature enables the system to generate event requests or wake up from STOP mode. Each external line can be configured independently to detect trigger events such as rising or falling edges, or both edges, as well as level events such as Low or High.

1.4.9 General-Purpose Input/Output (GPIO)

1.4.9.1 PCU Module

A34G43x series has a Port Control Unit (PCU) module that controls the external input and output (I/O) ports. By setting the PCU registers, users can configure the pins' uses, input/output direction, pull-up/pull-down, and debouncing for their applications as needed.

1.4.9.2 GPIO Module

Pins except the VDD, GND, and certain specific-purpose pins can be used as General-Purpose Input/Output (GPIO) pins.

The GPIO module controls the general I/O ports. Output pins can be set to generate high- or low-level signals by configuring the corresponding bits of the GPIO control register, while logic input pins can be monitored for their input status in the control registers.

1.4.10 Embedded Flash Memory

The flash memory controller is an interface controller for embedded flash memory. It manages data stored on the flash memory and communicates with other electronic devices.

The flash memory of A34G43x series has features as listed below:

- Code flash memory for 2048 KB with write protection bits
 - Dual Bank code flash (bank0: 1024 KB, bank1: 1024 KB) for OTA (Over-The-Air programming)
 - Bank swap with RWW (Read-While-Write)
- Data flash memory for 32 KB with write protection bits
- Erase units
 - Page (1 KB, code flash memory / 512 bytes, data flash memory)
 - Sector (4 KB, code flash memory / 2 KB, data flash memory)
 - Bulk (Up to 2048 KB, code flash memory / Up to 32 KB, data flash memory)
- Program unit: 2 Word (eight bytes), code flash memory / Word (four bytes), data flash memory
- Zero wait (less than 33 MHz), 1- to 15-wait, and cache (flash acceleration) access
- Read protection for code and data flash memories
- Data and instruction caches for code flash memory
- Built-in 16-bit specific polynomial for code and data flash memories
- ECC (Error Correction Code)
- Self-programming of the code flash memory
 - Supports updating data in a code flash memory region while executing user programs in the code flash area.

1.4.11 Direct Memory Access (DMA) Controller

The direct memory access (DMA) controller is used for high-speed data transfers between peripherals and memories. The DMA allows quick data transfers by copying or moving data between memory locations or between memory and peripherals, without involving the core.

- 16 channels
- 8-/ 16-/ 32-bit data transfers supported
- Various buffers with the same size supported
- DMA transfers are triggered through peripheral interrupts

1.4.12 Timers

1.4.12.1 16-Bit Timer

The A34G43x series has a TIMER module configured with 10 units. This 16-bit Timer supports four operating modes: Periodic Mode, PWM Mode, One-shot Mode, and Capture Mode.

Users can use a divided PCLK or an external clock as a clock source for the 16-bit Timer. An internal 10-bit prescaler allows to generate a variety of timer base clocks.

Interrupts can be triggered at regular intervals when the timer is used in Periodic Mode. Users can set the period and duty to form a PWM signal that is used in PWM Mode.

In One-shot Mode, the timer can generate one PWM waveform. The Capture Mode allows measuring the pulse intervals of an external input signal based on the predefined conditions. In addition, this timer can transmit signals to control other devices and is primarily used as a periodic tick timer or as a wake-up source.

1.4.12.2 Free-Run Timers (FRT)

A34G43x series has two built-in free-run timer (FRT) that are 32-bit up count timers. The FRT can run with an overflow interrupt or a match interrupt according to their register setting and can remain active in DEEP-SLEEP (STOP 1) mode.

1.4.12.3 Watchdog Timer (WDT)

The Watchdog Timer (WDT) is used to detect errors in the microcontroller caused by external interference or unexpected logical conditions. These errors cause the application program to deviate from its normal sequence. If the microcontroller loses control, the WDT will reset the microcontroller, allowing it to return to normal operation.

The WDT of the A34G43x series is a 32-bit down counter. If the WDT is set as a reset source, the microcontroller restarts when the down counter reaches 0.

When it is not used to monitor the microcontroller, the WDT can be used as a cycle timer along with an interrupt.

1.4.12.4 SysTick Timer

Although the SysTick timer is primarily designed for real-time operating systems, it can also be utilized as a standard down-counter.

The SysTick timer has the following features:

- 24-bit down-counter
- Automatic refreshing
- Maskable system interrupt generation when counter reaches zero.
- Programmable clock source

1.4.13 Pulse Width Modulation (PWM)

The A34G43x series has ten built-in PWM (Pulse-Width Modulation) generators. The PWM generators can output PWM (Pulse-Width Modulation) waveforms. This PWM (Pulse-Width Modulation) generator has a 16-bit resolution, five channels form a 1-unit, and each unit has a built-in prescaler. It can be used as a square waveform signal required for user application such as LED, motor, inverter, etc. through PWM related register setting.

1.4.14 Universal Asynchronous Receiver/Transmitter (UART)

The A34G43x series has a thirteen-channel UART module. The built-in UART module allows users to specify settings for transmitting and receiving data and provides the ability to read the current status of the UART. By providing information about the current state of transmission and reception processes, including types and conditions, the UART module enables users to monitor potential errors such as parity, overrun, framing, or break interrupts during data reception.

The UART module includes a programmable baud-rate generator for each channel, which generates an internal clock for its corresponding UART unit. This is achieved by dividing a prescaled clock using a baud-rate divisor that ranges from 1 to 65535 and then dividing the result by 16.

Users can program interrupts that can control UART communication.

Furthermore, the built-in DMA allows the UART module to transfer data directly to or from memory without involving the CPU, reducing CPU usage, and increasing the speed of data transfer.

Using receive FIFO, users can perform high-speed receive data.

1.4.15 Serial Peripheral Interface (SPI)

The A34G43x series has twelve built-in Serial Peripheral Interface (SPI) modules. The SPI modules are synchronized by clocks, and the specifications of the transfer clocks are adjustable.

The SPI module supports communication between one master and slaves. Slaves can be selected by the Slave Select (SS) signal.

The SPI module performs four-wire synchronous transfers via four signal terminals (SS, SCK, MOSI, and MISO). Its separate Transmit and Receive Buffers enable full-duplex communication, which is capable of reading and writing data simultaneously.

Specifically, the SPI module of the A34G43x series can enhance the Port Drive Strength so that the SPI rate can be supported up to 12 MHz with the enhanced Port Drive Strength.

Using receive FIFO, users can perform high-speed receive data.

1.4.16 Inter-Integrated Circuit (I2C) Interface

The I2C (Inter-Integrated Circuit) interface built in the A34G43x series satisfies the standard I2C communication protocol and is used for serial communication with internal and external devices via the I2C protocol.

Equipped with four units, it supports both master and slave modes and is capable of transmitting and receiving data in bytes by using interrupts or polling.

The I2C of the A34G43x series operates in Standard mode (100 kHz) or Fast mode (400 kHz) and supports General call.

It helps communicate with various peripherals that have the same bus type. To use the I2C, it is recommended to set the SCL and SDA pins to open-drain and then connect external pull-up resistors to render their output signals 'HIGH'.

Table 7. Features of I2C

I2C Features	I2C0	I2C1	I2C2	I2C3
7-bit addressing mode	O	O	O	O
Standard-mode (up to 100 kbit/s)	O	O	O	O
Fast-mode (up to 400 kbit/s)	O	O	O	O
General call	O	O	O	O

1.4.17 Motor Pulse-Width Modulation (MPWM)

The A34G43x series supports two Motor Pulse Width Modulation (MPWM) units. The MPWM in the A34G43x operates a total of three PWM modes as shown below to support various applications:

- Motor PWM mode for motor control
 - Supports up-down count and up-count PWM mode
- Normal PWM mode for a variety of PWM applications

In Motor or Normal PWM mode, the MPWM module of the A34G43x many different PWM control functionalities such as up to six channels' outputs, dead-time, up to six ADC triggers, a variety of interrupt event generations in many different conditions, protection events, and so on.

1.4.18 CAN With Flexible Data Rate (CANFD)

The A34G43x series includes two built-in Controller Area Network with Flexible Data Rate (CANFD) modules.

The CANFD module transmits and receives data according to the specified configuration and reads the current CANFD status. The CANFD protocol can be used for frame transmission (communication objects) and error handling management between network nodes.

The CANFD module supports CAN protocols, including CAN CC (Classic CAN 2.0B) and CAN FD (ISO 11898-1:2015). Interrupts for controlling CANFD communication can be programmed as needed.

1.4.19 Quad Serial Peripheral Interface (QSPI)

The QSPI (Quad Serial Peripheral Interface) built in the A34G43x series is a specialized communication interface targeting single, dual or quad QSPI NOR flash memories. It can operate in any of the two following modes:

- Indirect mode: all the operations are performed using the QUADSPI registers.
- memory-mapped mode: the external flash memory is mapped to the device address space and is seen by the system as if it was an internal memory.

1.4.20 Analog-to-Digital Converter (ADC)

The A34G43x series features a 12-bit resolution Analog-to-Digital Converter (ADC) module. It consists of three independent controllable units and supports up to 24 different analog inputs.

The ADC module supports A/D conversion of multiple channels in different modes, including single mode, sequential mode, multiple mode, inject mode and burst mode. It supports the sampling of voltage inputs from multiple channels at critical times, which is essential for various applications. This can be accomplished using the triggering functionality in conjunction with TIMER and MPWM.

Once the A/D conversion is complete, the result is stored in 16-bit data registers with left alignment.

The ADC is mapped to the AHB and capable of high-speed operation based on the internal clock source. In addition, it uses built-in DMA to reduce microcontroller core load while enabling fast data sampling.

The analog input monitoring and comparison capabilities of the ADC allow the applications to determine whether the input voltage is above or below the predefined level.

1.4.21 Digital-to-Analog Converter (DAC)

The Digital-to-Analog Converter (DAC) in A34G43x series provides a function to convert 12-bit digital value to analog output level. The DACxO pin can be used as general-purpose input/output (GPIO) when the DAC output is disconnected from the output pad and connected to on chip peripheral.

1.4.22 Comparator (CMP)

A34G43x has four comparators. Each comparator outputs the comparison results of two analog voltages via an I/O pin or generates interrupts.

Users can control the comparators by configuring the CMPn_CONF register. The HYSSEL[1:0] in CMPn_CONF register is used to select the comparator hysteresis voltage, and the HYSEN enables the hysteresis.

1.4.23 Advanced Encryption Standard (AES-128)

A34G43x series has a built-in Advanced Encryption Standard (AES-128) module that is used to encrypt or decrypt data.

The AES-128 module supports DMA single transfer for the incoming and outgoing data through 32-bit AMBA APB. (Two DMA channels are required.)

As a slave peripheral on the AMBA APB, the AES-128 module allows only single access of 32-bit word size. Writing other than 32-bit word size may corrupt the register content.

1.4.24 Random Number Generator (RNG)

The A34G43x series has a built-in Random Number Generator (RNG) that operates with an RNG clock, which users can select the clock source.

The RNG generates a random number based on the set seed value. If the users read data before the random number generation is completed, an error interrupt is flagged to prevent the users from reading the false data.

1.4.25 Cyclic Redundancy Check Calculation Module (CRC)

A Cyclic Redundancy Check (CRC) module is used to get 32-bit, 16-bit USB, 16-bit CCITT and 8-bit CCITT codes from 8-bit or 32-bit data size. Application programs employ CRC-based technologies to examine the integrity of data transfers, storages, and flash memories in conformance with functional safety standards.

Using this module, it is possible to examine the integrity of transfer data and flash memory data.

Specifically, the CRC module of the A34G43x series supports DMA.

1.5 Development Tools

In this chapter, various development tools for the A34G43x series are described. ABOV provides software tools, debuggers, and programmers to assist users in achieving the desired results for their target applications. ABOV supports the development ecosystem for our customers.

1.5.1 Compiler

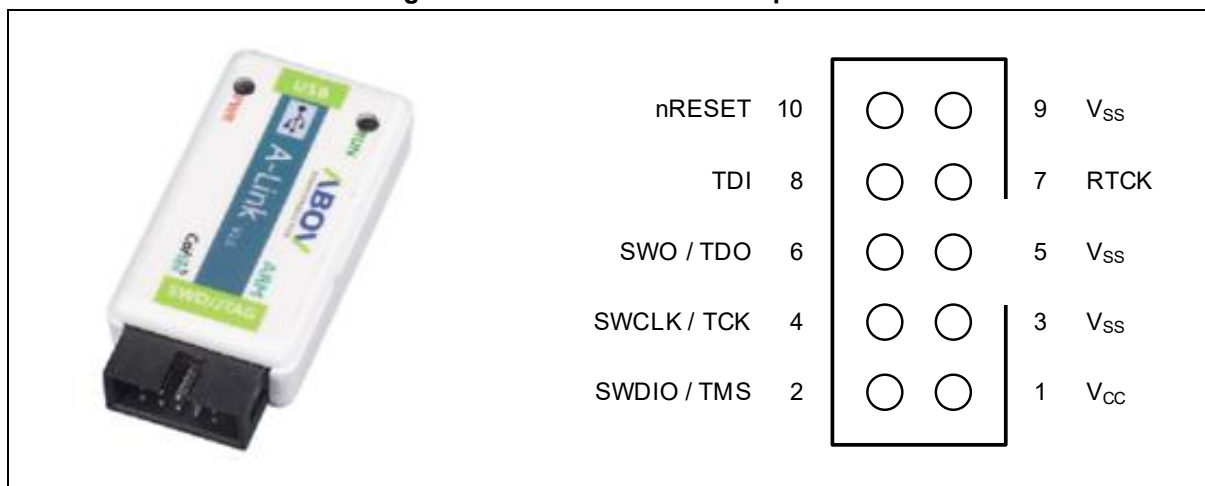
Since the A34G43x series has the Arm's high-performance 32-bit Cortex-M4F core for its CPU, users can use all kinds of third party's standard compilers such as Keil and IAR C Compiler.

1.5.2 Debugger

The A-Link and A-Link Pro can be used to emulate the A34G43x microcontroller using the SWD interface. The A-Link and A-Link Pro use a two-wire interface to connect the host computer to the microcontroller, which is connected to the user's system. The A-Link and A-Link Pro can read or change the value of microcontroller's internal memory and I/O peripherals. In addition, the A-Link and A-Link Pro control microcontroller's internal debugging logic. It means A-Link and A-Link Pro control emulation, step run, monitoring, and other functions regarding debugging.

Programming information using the A-Link and A-Link Pro is provided in Figure 2. For more detailed information about the A-Link and A-Link Pro, visit our official website and download the debugger software and related documents.

Figure 2. A-Link and Pin Description



1.5.3 Programmer

1.5.3.1 E-PGM+ and E-PGM Serial

E-PGM+ and E-PGM Serial are universal programmers and allow a user to program on the device directly.

- Supports for all ABOV microcontroller devices
- A dedicated tool for mass production
- 40-pin Textool DIP socket for single-chip programming (E-PGM+ only)
- 10-pin connector for ISP mode
- USB host interface
- HEX downloads and controls

1.5.3.2 Gang Programmer

E-Gang4 and E-Gang6 are capable of programming multiple devices simultaneously and can operate in both host-controlled and standalone modes without requiring a host computer connection. These programmers feature a USB interface for easy connection to a handler.

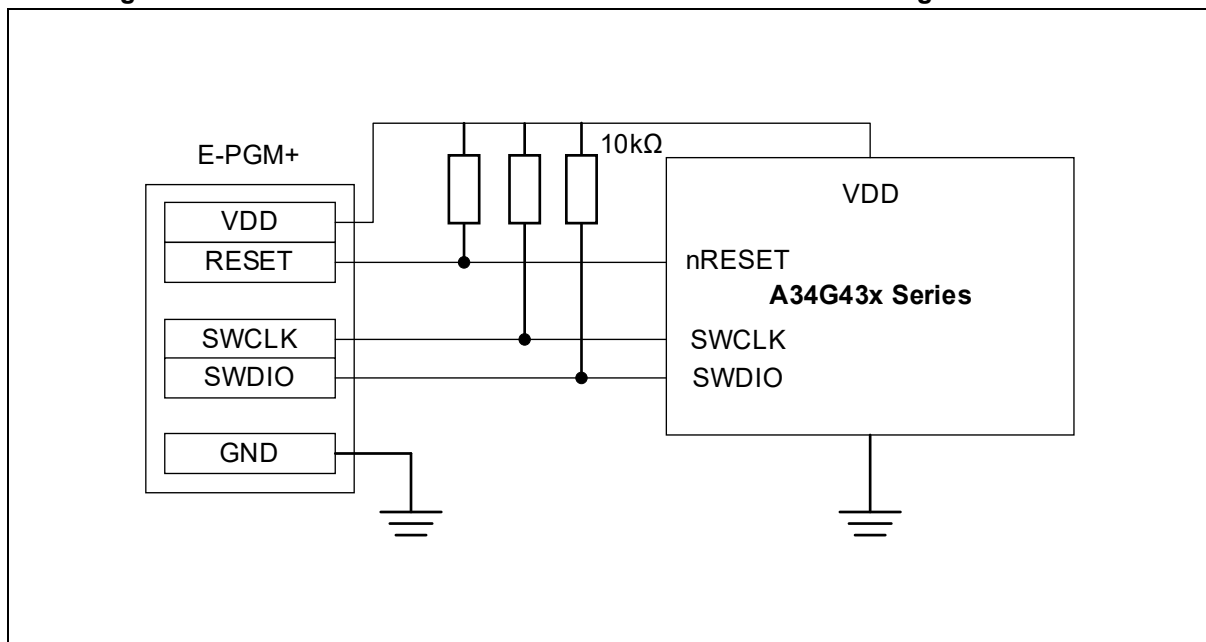
NOTE:

1. Refer to "Tools & Support > Programmer" on the [ABOV website](#).

1.5.3.3 SWD Debug Mode and E-PGM+ Connection

The connections between the A34G43x series and SWD debugger interface is illustrated in Figure 3.

Figure 3. Connection Between A34G43x Series and E-PGM+ Using SWD Interface



2. Pinouts and Pin Descriptions

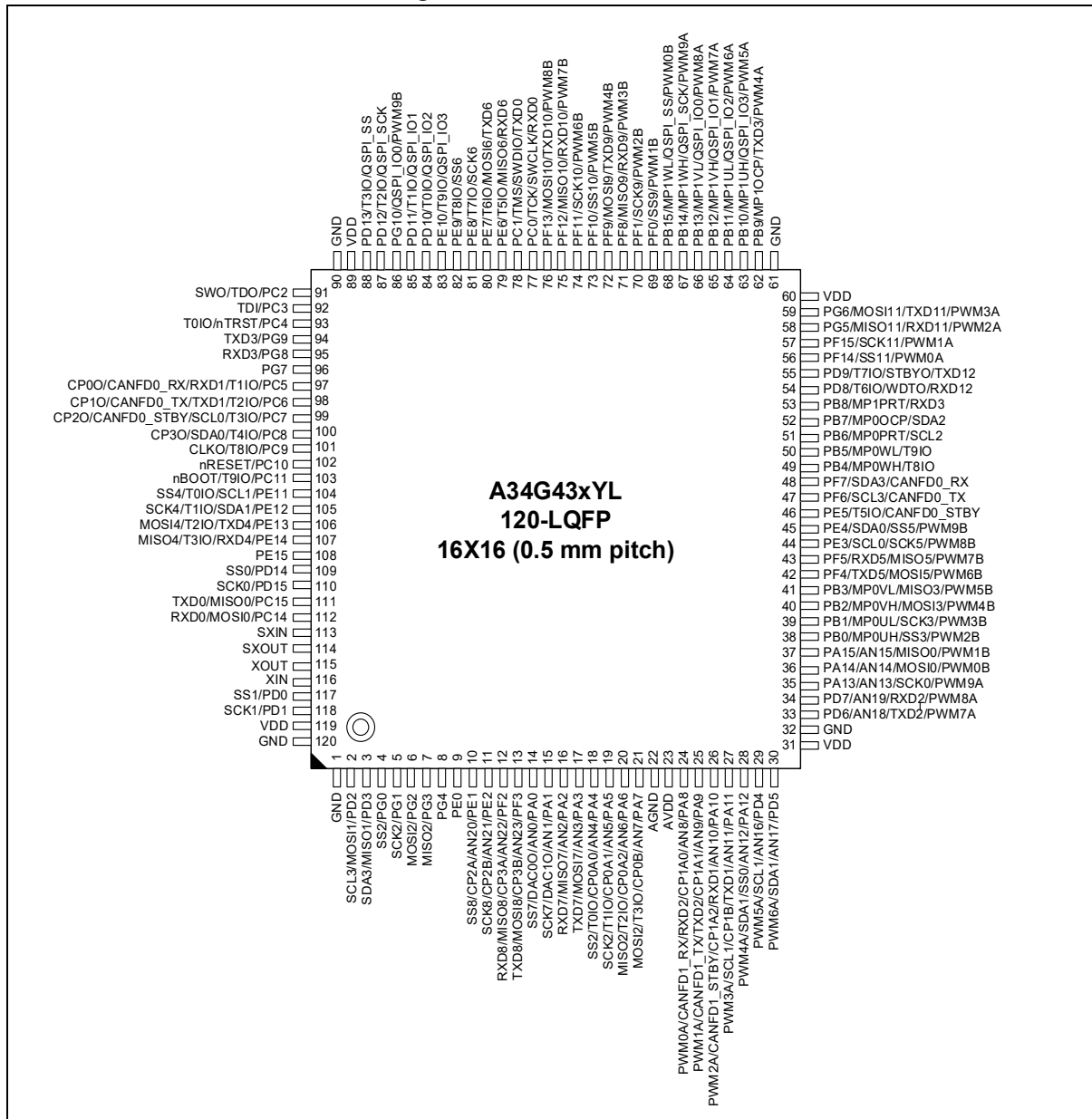
In this chapter, pinouts and pin descriptions of the A34G43x series are described.

2.1 Pinouts

Figure 4, Figure 5, and Figure 6 show the top views of the packages.

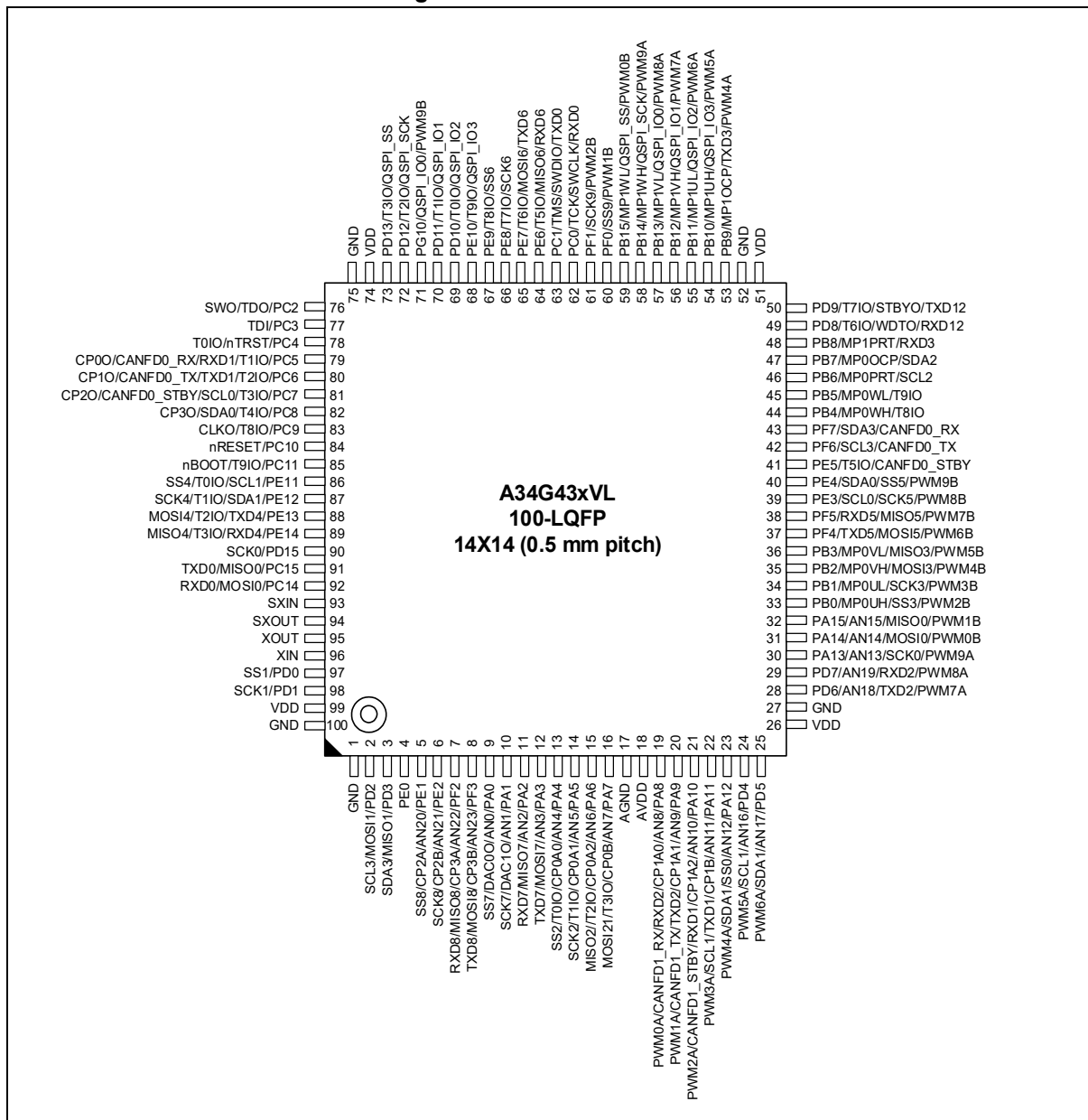
2.1.1 A34G43xYL (120-LQFP)

Figure 4. 120-LQFP Pinouts



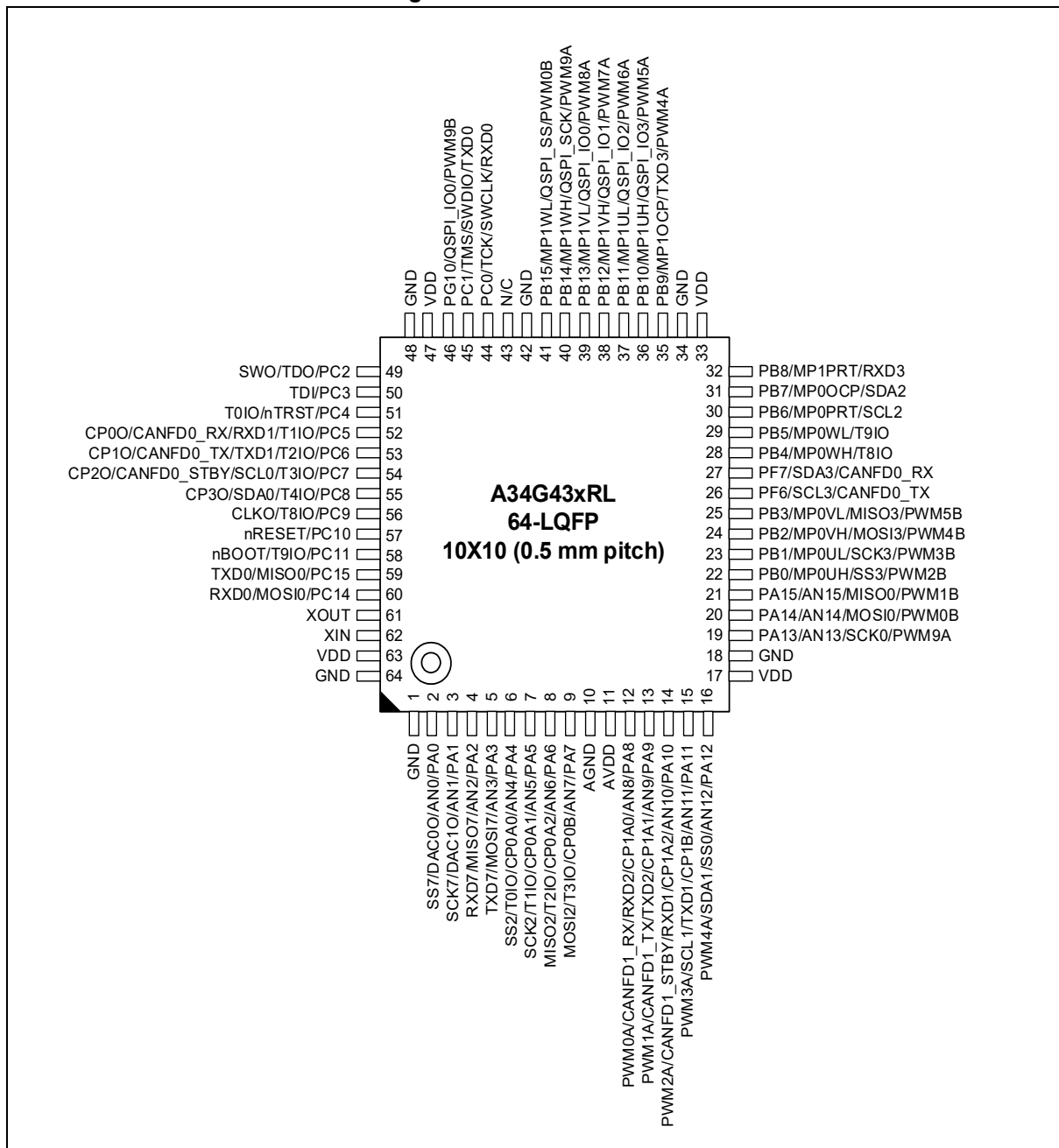
2.1.2 A34G43xVL (100-LQFP)

Figure 5. 100-LQFP Pinouts



2.1.3 A34G43xRL (64-LQFP)

Figure 6. 64-LQFP Pinouts



2.2 Pin Description

Table 8. Legend and Abbreviations used in Pin Description

Classification	Description	
Pin name	The pin name is always the same. Users can set a function on the pin using the alternate function.	
Classification	Abbreviation	Definition
Pin type	I	Input
	O	Output
	U	Pull-up
	D	Pull-down
	S	Schmitt-Trigger Input Type
	C	CMOS Input Type
	A	Analog
	P	Power
Pin function	Alternate functions	Functions selected through Pn_MR1/2 (PORT n MUX1/2 select register) registers

NOTES:

1. After a reset, all the pins are configured to function defined by their initial values. The initial value of the pin depends on the package type. This configuration follows the 120-pin standard.
2. nBOOT, nRESET, PC1 (SWDIO), PC0 (SWCLK), PC3 and PC4 are the default pull-up pins.
3. Do not configure unused pins as floating inputs. (low output is recommended).
4. After a reset, the internal pull-up for the boot pin is enabled.
5. After a reset, the internal pull-up for the serial wire clock (SWCLK) and the serial wire data I/O (SWDIO) is enabled.
6. The SWCLK and SWDIO pins should not be switched to other functions while they are being used.
7. PA11, PA12, PB6, PB7, PC7, PC8, PD2, PD3, PD4, PD5, PE3, PE4, PE11, PE12, PF6 and PF7 pins are configured as open-drain ports when used as I2C ports.
8. XIN, XOUT, SXOUT, and SXIN cannot be used for GPIO or other alternate functions. When the HSE and LSE clocks are not in use, the clocks must be disabled by using software.

Pin configuration information in Table 9 contains two pairs of power/ground and other dedicated pins. These multi-function pins have up to six selections of functions including GPIO. The Configuration including pin ordering can be changed without notice.

Table 9. Pin Description

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
1	1	1	GND	P	Ground	
2	2	-	PD2	IOUS	PORT D bit 2 input/output	
			MOSI1	IO	SPI channel 1 MOSI (master out/slave in) signal	
			SCL3 ⁽⁷⁾	IO	I2C channel 3 SCL input/output	Open-drain
3	3	-	PD3	IOUS	PORT D bit 3 input/output	
			MISO1	IO	SPI channel 1 MISO (master in/slave out) signal	
			SDA3 ⁽⁷⁾	IO	I2C channel 3 SDA input/output	Open-drain
4	-	-	PG0	IOUS	PORT G bit 0 input/output	
			SS2	IO	SPI channel 2 select signal input/output	
5	-	-	PG1	IOUS	PORT G bit 1 input/output	
			SCK2	IO	SPI channel 2 clock signal input/output	
6	-	-	PG2	IOUS	PORT G bit 2 input/output	
			MOSI2	IO	SPI channel 2 MOSI (master out/slave in) signal	
7	-	-	PG3	IOUS	PORT G bit 3 input/output	
			MISO2	IO	SPI channel 2 MISO (master in/slave out) signal	
8	-	-	PG4	IOUS	PORT G bit 4 input/output	
9	4	-	PE0	IOUS	PORT E bit 0 input/output	
10	5	-	PE1	IOUS	PORT E bit 1 input/output	
			SS8	IO	SPI channel 8 select signal input/output	
			AN20 ⁽¹⁾	IA	Analog input 20	
			CP2A ⁽¹⁾	IA	Comparator 2 input A	
11	6	-	PE2	IOUS	PORT E bit 2 input/output	
			SCK8	IO	SPI channel 8 clock signal input/output	
			AN21 ⁽¹⁾	IA	Analog input 21	
			CP2B ⁽¹⁾	IA	Comparator 2 input B	
12	7	-	PF2	IOUS	PORT F bit 2 input/output	
			RXD8	I	UART channel 8 RXD input	
			MISO8	IO	SPI channel 8 MISO (master in/slave out) signal	
			AN22 ⁽¹⁾	IA	Analog input 22	
			CP3A ⁽¹⁾	IA	Comparator 3 input A	
13	8	-	PF3	IOUS	PORT F bit 3 input/output	
			TXD8	O	UART channel 8 TXD output	
			MOSI8	IO	SPI channel 8 MOSI (master out/slave in) signal	
			AN23 ⁽¹⁾	IA	Analog input 23	
			CP3B ⁽¹⁾	IA	Comparator 3 input B	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
14	9	2	PA0	IOUS	PORT A bit 0 input/output	
			SS7	IO	SPI channel 7 select signal input/output	
			DAC0O	OA	Digital to analog 0 output	
			AN0 ⁽¹⁾	IA	Analog input 0	
15	10	3	PA1	IOUS	PORT A bit 1 input/output	
			SCK7	IO	SPI channel 7 clock signal input/output	
			DAC1O	OA	Digital to analog 1 output	
			AN1 ⁽¹⁾	IA	Analog input 1	
16	11	4	PA2	IOUS	PORT A bit 2 input/output	
			RXD7	Input	UART channel 7 RXD input	
			MISO7	IO	SPI channel 7 MISO (master in/slave out) signal	
			AN2 ⁽¹⁾	IA	Analog input 2	
17	12	5	PA3	IOUS	PORT A bit 3 input/output	
			TXD7	Output	UART channel 7 TXD output	
			MOSI7	IO	SPI channel 7 MOSI (master out/slave in) signal	
			AN3 ⁽¹⁾	IA	Analog input 3	
18	13	6	PA4	IOUS	PORT A bit 4 input/output	
			SS2	IO	SPI channel 2 select signal input/output	
			T0IO	IO	Timer 0 input/output	
			AN4 ⁽¹⁾	IA	Analog input 4	
			CP0A0 ⁽¹⁾	IA	Comparator 0 input A0	
19	14	7	PA5	IOUS	PORT A bit 5 input/output	
			SCK2	IO	SPI channel 2 clock signal input/output	
			T1IO	IO	Timer 1 input/output	
			AN5 ⁽¹⁾	IA	Analog input 5	
			CP0A1 ⁽¹⁾	IA	Comparator 0 input A1	
20	15	8	PA6	IOUS	PORT A bit 6 input/output	
			MISO2	IO	SPI channel 2 MISO (master in/slave out) signal	
			T2IO	IO	Timer 2 input/output	
			AN6 ⁽¹⁾	IA	Analog input 6	
			CP0A2 ⁽¹⁾	IA	Comparator 0 input A2	
21	16	9	PA7	IOUS	PORT A bit 7 input/output	
			MOSI2	IO	SPI channel 2 MOSI (master out/slave in) signal	
			T3IO	IO	Timer 3 input/output	
			AN7 ⁽¹⁾	IA	Analog input 7	
			CP0B ⁽¹⁾	IA	Comparator 0 input B	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
22	17	10	AGND	P	Analog Ground	
23	18	11	AVDD	P	Analog VDD	
24	19	12	PA8	IOUS	PORT A bit 8 input/output	
			RXD2	I	UART channel 2 RXD input	
			PWM0A	O	PWM channel 0 output	
			CANFD1_RX	I	CANFD channel 1 RXD input	
			AN8 ⁽¹⁾	IA	Analog input 8	
			CP1A0 ⁽¹⁾	IA	Comparator 1 input A0	
25	20	13	PA9	IOUS	PORT A bit 9 input/output	
			TXD2	O	UART channel 2 TXD output	
			PWM1A	O	PWM channel 1 output	
			CANFD1_TX	O	CANFD channel 1 TXD output	
			AN9 ⁽¹⁾	IA	Analog input 9	
			CP1A1 ⁽¹⁾	IA	Comparator 1 input A1	
26	21	14	PA10	IOUS	PORT A bit 10 input/output	
			RXD1	I	UART channel 1 RXD input	
			PWM2A	O	PWM channel 2 output	
			CANFD1_STBY	O	CANFD channel 1 standby signal output	
			AN10 ⁽¹⁾	IA	Analog input 10	
			CP1A2 ⁽¹⁾	IA	Comparator 1 input A2	
27	22	15	PA11	IOUS	PORT A bit 11 input/output	
			TXD1	O	UART channel 1 TXD output	
			PWM3A	O	PWM channel 3 output	
			AN11 ⁽¹⁾	IA	Analog input 11	
			CP1B ⁽¹⁾	IA	Comparator 1 input B	
			SCL1 ⁽⁷⁾	IO	I2C channel 1 SCL input/output	Open-drain
28	23	16	PA12	IOUS	PORT A bit 12 input/output	
			SS0	IO	SPI channel 0 select signal input/output	
			PWM4A	O	PWM channel 4 output	
			AN12 ⁽¹⁾	IA	Analog input 12	
			SDA1 ⁽⁷⁾	IO	I2C channel 1 SDA input/output	Open-drain
29	24	-	PD4	IOUS	PORT D bit 4 input/output	
			PWM5A	O	PWM channel 5 output	
			AN16 ⁽¹⁾	IA	Analog input 16	
			SCL1 ⁽⁷⁾	IO	I2C channel 1 SCL input/output	Open-drain
30	25	-	PD5	IOUS	PORT D bit 5 input/output	
			PWM6A	O	PWM channel 6 output	
			AN17 ⁽¹⁾	IA	Analog input 17	
			SDA1 ⁽⁷⁾	IO	I2C channel 1 SDA input/output	Open-drain
31	26	17	VDD	P	VDD	
32	27	18	GND	P	Ground	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
33	28	-	PD6	IOUS	PORT D bit 6 input/output	
			PWM7A	O	PWM channel 7 output	
			AN18 ⁽¹⁾	IA	Analog input 18	
			TXD2	O	UART channel 2 TXD output	
34	29	-	PD7	IOUS	PORT D bit 7 input/output	
			PWM8A	O	PWM channel 8 output	
			AN19 ⁽¹⁾	IA	Analog input 19	
			RXD2	I	UART channel 2 RXD input	
35	30	19	PA13	IOUS	PORT A bit 13 input/output	
			PWM9A	O	PWM channel 9 output	
			AN13 ⁽¹⁾	IA	Analog input 13	
			SCK0	IO	SPI channel 0 clock input/output	
36	31	20	PA14	IOUS	PORT A bit 14 input/output	
			PWM0B	O	PWM channel 0 inversion output	
			AN14 ⁽¹⁾	IA	Analog input 14	
			MOSI0	IO	SPI channel 0 MOSI (master out/slave in) signal	
37	32	21	PA15	IOUS	PORT A bit 15 input/output	
			PWM1B	O	PWM channel 1 inversion output	
			AN15 ⁽¹⁾	IA	Analog input 15	
			MISO0	IO	SPI channel 0 MISO (master in/slave out) signal	
38	33	22	PB0	IOUS	PORT B bit 0 input/output	
			SS3	IO	SPI channel 3 select signal input/output	
			PWM2B	O	PWM channel 2 inversion output	
			MP0UH	O	MPWM channel 0 UH output	
39	34	23	PB1	IOUS	PORT B bit 1 input/output	
			SCK3	IO	SPI channel 3 clock input/output	
			PWM3B	O	PWM channel 3 inversion output	
			MP0UL	O	MPWM channel 0 UL output	
40	35	24	PB2	IOUS	PORT B bit 2 input/output	
			MOSI3	IO	SPI channel 3 MOSI (master out/slave in) signal	
			PWM4B	O	PWM channel 4 inversion output	
			MP0VH	O	MPWM channel 0 VH output	
41	36	25	PB3	IOUS	PORT B bit 3 input/output	
			MISO3	IO	SPI channel 3 MISO (master in/slave out) signal	
			PWM5B	O	PWM channel 5 inversion output	
			MP0VL	O	MPWM channel 0 VL output	
42	37	-	PF4	IOUS	PORT F bit 4 input/output	
			TXD5	O	UART channel 5 TXD output	
			PWM6B	O	PWM channel 6 inversion output	
			MOSI5	IO	SPI channel 5 MOSI (master out/slave in) signal	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
43	38	-	PF5	IOUS	PORT F bit 5 input/output	
			RXD5	I	UART channel 5 RXD input	
			PWM7B	O	PWM channel 7 inversion output	
			MISO5	IO	SPI channel 5 MISO (master in/slave out) signal	
44	39	-	PE3	IOUS	PORT E bit 3 input/output	
			PWM8B	O	PWM channel 8 inversion output	
			SCK5	IO	SPI channel 5 clock input/output	
			SCL0 ⁽⁷⁾	IO	I2C channel 0 SCL input/output	Open-drain
45	40	-	PE4	IOUS	PORT E bit 4 input/output	
			PWM9B	O	PWM channel 9 inversion output	
			SS5	IO	SPI channel 5 select signal input/output	
			SDA0 ⁽⁷⁾	IO	I2C channel 0 SDA input/output	Open-drain
46	41	-	PE5	IOUS	PORT E bit 5 input/output	
			T5IO	IO	Timer 5 input/output	
			CANFD0_STBY	O	CANFD channel 0 standby signal output	
47	42	26	PF6	IOUS	PORT F bit 6 input/output	
			CANFD0_TX	O	CANFD channel 0 TXD output	
			SCL3 ⁽⁷⁾	IO	I2C channel 3 SCL input/output	Open-drain
48	43	27	PF7	IOUS	PORT F bit 7 input/output	
			CANFD0_RX	I	CANFD channel 0 RXD input	
			SDA3 ⁽⁷⁾	IO	I2C channel 3 SDA input/output	Open-drain
49	44	28	PB4	IOUS	PORT B bit 4 input/output	
			MP0WH	O	MPWM channel 0 WH output	
			T8IO	IO	Timer 8 input/output	
50	45	29	PB5	IOUS	PORT B bit 5 input/output	
			MP0WL	O	MPWM channel 0 WL output	
			T9IO	IO	Timer 9 input/output	
51	46	30	PB6	IOUS	PORT B bit 6 input/output	
			MP0PRT	I	MPWM0 Protection input Signal	
			SCL2 ⁽⁷⁾	IO	I2C channel 2 SCL input/output	Open-drain
52	47	31	PB7	IOUS	PORT B bit 7 input/output	
			MP0OCP	I	MPWM0 Over-current protection input Signal	
			SDA2 ⁽⁷⁾	IO	I2C channel 2 SDA input/output	Open-drain
53	48	32	PB8	IOUS	PORT B bit 8 input/output	
			RXD3	I	UART channel 3 RXD input	
			MP1PRT	I	MPWM1 Protection input Signal	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
54	49	-	PD8	IOUS	PORT D bit 8 input/output	
			RXD12	I	UART channel 12 RXD input	
			T6IO	IO	Timer 6 input/output	
			WDTO	O	WDT output	
55	50	-	PD9	IOUS	PORT D bit 9 input/output	
			TXD12	I	UART channel 12 TXD output	
			T7IO	IO	Timer 7 input/output	
			STBYO	O	Power-down Mode Indication Signal	
56	-	-	PF14	IOUS	PORT F bit 14 input/output	
			PWM0A	O	PWM channel 0 output	
			SS11	IO	SPI channel 11 select signal input/output	
57	-	-	PF15	IOUS	PORT F bit 15 input/output	
			PWM1A	O	PWM channel 1 output	
			SCK11	IO	SPI channel 11 clock input/output	
58	-	-	PG5	IOUS	PORT G bit 5 input/output	
			RXD11	I	UART channel 11 RXD input	
			PWM2A	O	PWM channel 2 output	
			MISO11	IO	SPI channel 11 MISO (master in/slave out) signal	
59	-	-	PG6	IOUS	PORT G bit 6 input/output	
			TXD11	O	UART channel 11 TXD output	
			PWM3A	O	PWM channel 3 output	
			MOSI11	IO	SPI channel 11 MOSI (master out/slave in) signal	
60	51	33	VDD	P	VDD	
61	52	34	GND	P	Ground	
62	53	35	PB9	IOUS	PORT B bit 9 input/output	
			TXD3	O	UART channel 3 TXD output	
			PWM4A	O	PWM channel 4 output	
			MP1OCP	I	MPWM1 Over-current protection input Signal	
63	54	36	PB10	IOUS	PORT B bit 10 input/output	
			QSPI_IO3	IO	QSPI channel 0 IO3 input/output	
			PWM5A	O	PWM channel 5 output	
			MP1UH	O	MPWM channel 1 UH output	
64	55	37	PB11	IOUS	PORT B bit 11 input/output	
			QSPI_IO2	IO	QSPI channel 0 IO2 input/output	
			PWM6A	O	PWM channel 6 output	
			MP1UL	O	MPWM channel 1 UL output	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
65	56	38	PB12	IOUS	PORT B bit 12 input/output	
			QSPI_IO1	IO	QSPI channel 0 IO1 input/output	
			PWM7A	O	PWM channel 7 output	
			MP1VH	O	MPWM channel 1 VH output	
66	57	39	PB13	IOUS	PORT B bit 13 input/output	
			QSPI_IO0	IO	QSPI channel 0 IO0 input/output	
			PWM8A	O	PWM channel 8 output	
			MP1VL	O	MPWM channel 1 VL output	
67	58	40	PB14	IOUS	PORT B bit 14 input/output	
			QSPI_SCK	IO	QSPI channel 0 clock input/output	
			PWM9A	O	PWM channel 9 output	
			MP1WH	O	MPWM channel 1 WH output	
68	59	41	PB15	IOUS	PORT B bit 15 input/output	
			QSPI_SS	IO	QSPI channel 0 select signal input/output	
			PWM0B	O	PWM channel 0 inversion output	
			MP1WL	O	MPWM channel 1 WL output	
69	60	-	PF0	IOUS	PORT F bit 0 input/output	
			PWM1B	O	PWM channel 1 inversion output	
			SS9	IO	SPI channel 9 select signal input/output	
70	61	-	PF1	IOUS	PORT F bit 1 input/output	
			PWM2B	O	PWM channel 2 inversion output	
			SCK9	IO	SPI channel 9 clock input/output	
71	-	-	PF8	IOUS	PORT F bit 8 input/output	
			RXD9	I	UART channel 9 RXD input	
			PWM3B	O	PWM channel 3 inversion output	
			MISO9	IO	SPI channel 9 MISO (master in/slave out) signal	
72	-	-	PF9	IOUS	PORT F bit 9 input/output	
			TXD9	O	UART channel 9 TXD output	
			PWM4B	O	PWM channel 4 inversion output	
			MOSI9	IO	SPI channel 9 MOSI (master out/slave in) signal	
73	-	-	PF10	IOUS	PORT F bit 10 input/output	
			PWM5B	O	PWM channel 5 inversion output	
			SS10	IO	SPI channel 10 select signal input/output	
74	-	-	PF11	IOUS	PORT F bit 11 input/output	
			PWM6B	O	PWM channel 6 inversion output	
			SCK10	IO	SPI channel 10 clock input/output	
75	-	-	PF12	IOUS	PORT F bit 12 input/output	
			RXD10	I	UART channel 10 RXD input	
			PWM7B	O	PWM channel 7 inversion output	
			MISO10	IO	SPI channel 10 MISO (master in/slave out) signal	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type	Description	Remark
120-pin	100-pin	64-pin				
76	-	-	PF13	IOUS	PORT F bit 13 input/output	
			TXD10	O	UART channel 10 TXD output	
			PWM8B	O	PWM channel 8 inversion output	
			MOSI10	IO	SPI channel 10 MOSI (master out/slave in) signal	
-	-	42	GND	P	Ground	
-	-	43	N/C	-	N/C	
77	62	44	PC0	IOUS	PORT C bit 0 input/output	
			TCK / SWCLK ⁽¹⁾⁽²⁾⁽⁵⁾⁽⁶⁾	I	JTAG TCK, SWD clock input	Pull-up
			RXD0	I	UART channel 0 RXD input	
78	63	45	PC1	IOUS	PORT C bit 1 input/output	
			TMS / SWDIO ⁽¹⁾⁽²⁾⁽⁵⁾⁽⁶⁾	IO	JTAG TMS, SWD data input/output	Pull-up
			TXD0	O	UART channel 0 TXD output	
79	64	-	PE6	IOUS	PORT E bit 6 input/output	
			RXD6	I	UART channel 6 RXD input	
			T5IO	IO	Timer 5 input/output	
			MISO6	IO	SPI channel 6 MISO (master in/slave out) signal	
80	65	-	PE7	IOUS	PORT E bit 7 input/output	
			TXD6	O	UART channel 6 TXD output	
			T6IO	IO	Timer 6 input/output	
			MOSI6	IO	SPI channel 6 MOSI (master out/slave in) signal	
81	66	-	PE8	IOUS	PORT E bit 8 input/output	
			T7IO	IO	Timer 7 input/output	
			SCK6	IO	SPI channel 6 clock input/output	
82	67	-	PE9	IOUS	PORT E bit 9 input/output	
			T8IO	IO	Timer 8 input/output	
			SS6	IO	SPI channel 6 select signal input/output	
83	68	-	PE10	IOUS	PORT E bit 10 input/output	
			QSPI_IO3	IO	QSPI channel 0 IO3 input/output	
			T9IO	IO	Timer 9 input/output	
84	69	-	PD10	IOUS	PORT D bit 10 input/output	
			QSPI_IO2	IO	QSPI channel 0 IO2 input/output	
			T0IO	IO	Timer 0 input/output	
85	70	-	PD11	IOUS	PORT D bit 11 input/output	
			QSPI_IO1	IO	QSPI channel 0 IO1 input/output	
			T1IO	IO	Timer 1 input/output	
86	71	46	PG10	IOUS	PORT G bit 10 input/output	
			QSPI_IO0	IO	QSPI channel 0 IO0 input/output	
			PWM9B	O	PWM channel 9 inversion output	

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type ⁽¹⁾	Description	Remark
120-pin	100-pin	64-pin				
87	72	-	PD12	IOUS	PORT D bit 12 input/output	
			QSPI_SCK	IO	QSPI channel 0 clock input/output	
			T2IO	IO	Timer 2 input/output	
88	73	-	PD13	IOUS	PORT D bit 13 input/output	
			QSPI_SS	IO	QSPI channel 0 select signal input/output	
			T3IO	IO	Timer 3 input/output	
89	74	47	VDD	P	VDD	
90	75	48	GND	P	Ground	
91	76	49	PC2	IOUS	PORT C bit 2 input/output	
			TDO / SWO ⁽¹⁾	O	JTAG TDO output, Serial Wire Output	Output Low
92	77	50	PC3	IOUS	PORT C bit 3 input/output	
			TDI ⁽¹⁾	I	JTAG TDI input	Pull-up
93	78	51	PC4	IOUS	PORT C bit 4 input/output	
			nTRST ⁽¹⁾	O	JTAG nTRST input	Pull-up
			T0IO	IO	Timer 0 input/output	
94	-	-	PG9	IOUS	PORT G bit 9 input/output	
			TXD3	O	UART channel 3 TXD output	
95	-	-	PG8	IOUS	PORT G bit 8 input/output	
			RXD3	I	UART channel 3 RXD input	
96	-	-	PG7	IOUS	PORT G bit 7 input/output	
97	79	52	PC5	IOUS	PORT C bit 5 input/output	
			T1IO	IO	Timer 1 input/output	
			RXD1	I	UART channel 1 RXD input	
			CANFD0_RX	I	CANFD channel 0 RXD input	
			CP0O	OA	Comparator 0 output	
98	80	53	PC6	IOUS	PORT C bit 6 input/output	
			T2IO	IO	Timer 2 input/output	
			TXD1	O	UART channel 1 TXD output	
			CANFD0_TX	O	CANFD channel 0 TXD output	
			CP1O	OA	Comparator 1 output	
99	81	54	PC7	IOUS	PORT C bit 7 input/output	
			CP2O	OA	Comparator 2 output	
			T3IO	IO	Timer 3 input/output	
			SCL0 ⁽⁷⁾	IO	I2C channel 0 SCL input/output	Open-drain
			CANFD0_STBY	O	CANFD channel 0 standby signal output	
100	82	55	PC8	IOUS	PORT C bit 8 input/output	
			CP3O	OA	Comparator 3 output	
			T4IO	IO	Timer 4 input/output	
			SDA0 ⁽⁷⁾	IO	I2C channel 0 SDA input/output	Open-drain
101	83	56	PC9	IOUS	PORT C bit 9 input/output	
			T8IO	IO	Timer 8 input/output	
			CLKO	O	System Clock output	
102	84	57	PC10	IOUS	PORT C bit 9 input/output	
			T9IO	IO	Timer 9 input/output	
			nRESET ⁽¹⁾⁽²⁾	I	External Reset input	Pull-up

Table 9. Pin Description (continued)

Pin No.			Pin Name	Type ⁽¹⁾	Description	Remark
120-pin	100-pin	64-pin				
103	85	58	PC11	IOUS	PORT C bit 11 input/output	
			T9IO	IO	Timer 9 input/output	
			nBOOT ⁽¹⁾⁽²⁾⁽⁴⁾	I	Boot Mode Selection input	Pull-up
104	86	-	PE11	IOUS	PORT E bit 11 input/output	
			SS4	IO	SPI channel 4 select signal input/output	
			SCL1 ⁽⁷⁾	IO	I2C channel 1 SCL input/output	Open-drain
			T0IO	IO	Timer 0 input/output	
105	87	-	PE12	IOUS	PORT E bit 12 input/output	
			SCK4	IO	SPI channel 4 clock input/output	
			SDA1 ⁽⁷⁾	IO	I2C channel 1 SDA input/output	Open-drain
			T1IO	IO	Timer 1 input/output	
106	88	-	PE13	IOUS	PORT E bit 13 input/output	
			MOSI4	IO	SPI channel 4 MOSI (master out/slave in) signal	
			TXD4	O	UART channel 4 TXD output	
			T2IO	IO	Timer 2 input/output	
107	89	-	PE14	IOUS	PORT E bit 14 input/output	
			MISO4	IO	SPI channel 4 MISO (master in/slave out) signal	
			RXD4	I	UART channel 4 RXD input	
			T3IO	IO	Timer 3 input/output	
108	-	-	PE15	IOUS	PORT E bit 15 input/output	
109	-	-	PD14	IOUS	PORT D bit 14 input/output	
			SS0	IO	SPI channel 0 select signal input/output	
110	90	-	PD15	IOUS	PORT D bit 15 input/output	
			SCK0	IO	SPI channel 0 clock input/output	
111	91	59	PC15	IOUS	PORT C bit 15 input/output	
			MISO0	IO	SPI channel 0 MISO (master in/slave out) signal	
			TXD0	O	UART channel 0 TXD output	
112	92	60	PC14	IOUS	PORT C bit 14 input/output	
			MOSI0	IO	SPI channel 0 MOSI (master out/slave in) signal	
			RXD0	I	UART channel 0 RXD input	
113	93	-	SXIN ⁽¹⁾⁽⁸⁾	IA	Sub crystal oscillator input	
114	94	-	SXOUT ⁽¹⁾⁽⁸⁾	OA	Sub crystal oscillator output	
115	95	61	XOUT ⁽¹⁾⁽⁸⁾	OA	External crystal oscillator output	
116	96	62	XIN ⁽¹⁾⁽⁸⁾	IA	External crystal oscillator input	
117	97	-	PD0	IOUS	PORT D bit 0 input/output	
			SS1	IO	SPI channel 1 select signal input/output	
118	98	-	PD1	IOUS	PORT D bit 1 input/output	
			SCK1	IO	SPI channel 1 clock input/output	
119	99	63	VDD	P	VDD	
120	100	64	GND	P	Ground	

2.3 Alternate Function Pins

GPIO pins have alternate functions as described in Table 10.

Table 10. GPIO Alternate Function

Pin Name	Alternate Function					
	AF0	AF1	AF2	AF3	AF4	AF7
PA0	PA0			SS7		AN0 / DAC00 ⁽¹⁾
PA1	PA1			SCK7		AN1 / DAC10 ⁽¹⁾
PA2	PA2	RXD7		MISO7		AN2 ⁽¹⁾
PA3	PA3	TXD7		MOSI7		AN3 ⁽¹⁾
PA4	PA4	SS2	T0IO			AN4 / CP0A0 ⁽¹⁾
PA5	PA5	SCK2	T1IO			AN5 / CP0A1 ⁽¹⁾
PA6	PA6	MISO2	T2IO			AN6 / CP0A2 ⁽¹⁾
PA7	PA7	MOSI2	T3IO			AN7 / CP0B ⁽¹⁾
PA8	PA8	RXD2	PWM0A	CANFD1_RX		AN8 / CP1A0 ⁽¹⁾
PA9	PA9	TXD2	PWM1A	CANFD1_TX		AN9 / CP1A1 ⁽¹⁾
PA10	PA10	RXD1	PWM2A	CANFD1_STBY		AN10 / CP1A2 ⁽¹⁾
PA11	PA11	TXD1	PWM3A	SCL1		AN11 / CP1B ⁽¹⁾
PA12	PA12	SS0	PWM4A	SDA1		AN12 ⁽¹⁾
PA13	PA13	SCK0	PWM9A			AN13 ⁽¹⁾
PA14	PA14	MOSI0	PWM0B			AN14 ⁽¹⁾
PA15	PA15	MISO0	PWM1B			AN15 ⁽¹⁾
PB0	PB0	SS3	PWM2B	MP0UH		⁽¹⁾
PB1	PB1	SCK3	PWM3B	MP0UL		⁽¹⁾
PB2	PB2	MOSI3	PWM4B	MP0VH		⁽¹⁾
PB3	PB3	MISO3	PWM5B	MP0VL		⁽¹⁾
PB4	PB4		T8IO	MP0WH		⁽¹⁾
PB5	PB5		T9IO	MP0WL		⁽¹⁾
PB6	PB6	SCL2		MP0PRT		⁽¹⁾
PB7	PB7	SDA2		MP0OCP		⁽¹⁾
PB8	PB8	RXD3		MP1PRT		⁽¹⁾
PB9	PB9	TXD3	PWM4A	MP1OCP		⁽¹⁾
PB10	PB10	QSPI_IO3	PWM5A	MP1UH		⁽¹⁾
PB11	PB11	QSPI_IO2	PWM6A	MP1UL		⁽¹⁾
PB12	PB12	QSPI_IO1	PWM7A	MP1VH		⁽¹⁾
PB13	PB13	QSPI_IO0	PWM8A	MP1VL		⁽¹⁾
PB14	PB14	QSPI_SCK	PWM9A	MP1WH		⁽¹⁾
PB15	PB15	QSPI_SS	PWM0B	MP1WL		⁽¹⁾
PC0	PC0	RXD0		TCK / SWCLK ⁽¹⁾		
PC1	PC1	TXD0		TMS / SWDIO ⁽¹⁾		
PC2	PC2			TDO / SWO ⁽¹⁾		
PC3	PC3			TDI ⁽¹⁾		
PC4	PC4		T0IO	nTRST ⁽¹⁾		
PC5	PC5	RXD1	T1IO	CANFD0_RX	CP00	⁽¹⁾
PC6	PC6	TXD1	T2IO	CANFD0_TX	CP10	⁽¹⁾
PC7	PC7	SCL0	T3IO	CANFD0_STBY	CP20	⁽¹⁾
PC8	PC8	SDA0	T4IO		CP30	⁽¹⁾

Table 10. GPIO Alternate Function (continued)

Pin Name	Alternate Function					
	AF0	AF1	AF2	AF3	AF4	AF7
PC9	PC9		T8IO	CLKO		(1)
PC10	PC10			nRESET ⁽¹⁾		
PC11	PC11		T9IO	nBOOT ⁽¹⁾		
PC14	PC14	MOSI0		RXD0		
PC15	PC15	MISO0		TXD0		
PD0	PD0	SS1				(1)
PD1	PD1	SCK1				(1)
PD2	PD2	MOSI1		SCL3		(1)
PD3	PD3	MISO1		SDA3		(1)
PD4	PD4	SCL1	PWM5A			AN16 ⁽¹⁾
PD5	PD5	SDA1	PWM6A			AN17 ⁽¹⁾
PD6	PD6	TXD2	PWM7A			AN18 ⁽¹⁾
PD7	PD7	RXD2	PWM8A			AN19 ⁽¹⁾
PD8	PD8	RXD12	T6IO	WDTO		(1)
PD9	PD9	TXD12	T7IO	STBYO		(1)
PD10	PD10	QSPI_IO2	T0IO			(1)
PD11	PD11	QSPI_IO1	T1IO			(1)
PD12	PD12	QSPI_SCK	T2IO			(1)
PD13	PD13	QSPI_SS	T3IO			(1)
PD14	PD14	SS0				(1)
PD15	PD15	SCK0				(1)
PE0	PE0					(1)
PE1	PE1			SS8		AN20 / CP2A ⁽¹⁾
PE2	PE2			SCK8		AN21 / CP2B ⁽¹⁾
PE3	PE3	SCL0	PWM8B	SCK5		(1)
PE4	PE4	SDA0	PWM9B	SS5		(1)
PE5	PE5		T5IO	CANFD0_STBY		(1)
PE6	PE6	RXD6	T5IO	MISO6		(1)
PE7	PE7	TXD6	T6IO	MOSI6		(1)
PE8	PE8		T7IO	SCK6		(1)
PE9	PE9		T8IO	SS6		(1)
PE10	PE10	QSPI_IO3	T9IO			(1)
PE11	PE11	SCL1	T0IO	SS4		(1)
PE12	PE12	SDA1	T1IO	SCK4		(1)
PE13	PE13	TXD4	T2IO	MOSI4		(1)
PE14	PE14	RXD4	T3IO	MISO4		(1)
PE15	PE15					(1)
PF0	PF0		PWM1B	SS9		(1)
PF1	PF1		PWM2B	SCK9		(1)
PF2	PF2	RXD8		MISO8		AN22 / CP3A ⁽¹⁾
PF3	PF3	TXD8		MOSI8		AN23 / CP3B ⁽¹⁾
PF4	PF4	TXD5	PWM6B	MOSI5		(1)

Table 10. GPIO Alternate Function (continued)

Pin Name	Alternate Function					
	AF0	AF1	AF2	AF3	AF4	AF7
PF5	PF5	RXD5	PWM7B	MISO5		(1)
PF6	PF6	SCL3		CANFD0_TX		(1)
PF7	PF7	SDA3		CANFD0_RX		(1)
PF8	PF8	RXD9	PWM3B	MISO9		(1)
PF9	PF9	TXD9	PWM4B	MOSI9		(1)
PF10	PF10		PWM5B	SS10		(1)
PF11	PF11		PWM6B	SCK10		(1)
PF12	PF12	RXD10	PWM7B	MISO10		(1)
PF13	PF13	TXD10	PWM8B	MOSI10		(1)
PF14	PF14		PWM0A	SS11		(1)
PF15	PF15		PWM1A	SCK11		(1)
PG0	PG0	SS2				(1)
PG1	PG1	SCK2				(1)
PG2	PG2	MOSI2				(1)
PG3	PG3	MISO2				(1)
PG4	PG4					(1)
PG5	PG5	RXD11	PWM2A	MISO11		(1)
PG6	PG6	TXD11	PWM3A	MOSI11		(1)
PG7	PG7					(1)
PG8	PG8	RXD3				(1)
PG9	PG9	TXD3				(1)
PG10	PG10	QSPI_IO0	PWM9B			(1)
XIN						
XOUT						
SXIN						
SXOUT						

NOTES:

1. It means 'selected pin function after reset condition'. (The initial value of the pin is different depending on the package type)
2. Unused pins are set to output from firmware (low output is recommended).
3. PC12 and PC13 ports are not available. When configuring PC_MR2, PC_CR, PC_PRCR, PC_DER, PC_STR, PC_IER, PC_ISR, PC_ICR, PC_ODR, PC_IDR, PC_BSR, PC_BCR registers, the 'default values' must be maintained.

3. Electrical Characteristics

3.1 Parameter Conditions

Unless otherwise specified, all voltages are referenced to GND.

3.1.1 Minimum and Maximum Values

Unless otherwise specified, our production tests guarantee the minimum and maximum values of the device under the worst-case conditions including ambient temperature, supply voltage, and frequency.

Data based on characterization result, design simulation, and/or technical characteristics are not tested in production but are indicated in the table footnotes.

3.1.2 Typical Values

Unless otherwise specified, typical data are based on the conditions of $T_A = 25^{\circ}\text{C}$ and $V_{DD} = AV_{DD} = 5.0\text{ V}$. The typical data are provided only as design recommendations and are not tested.

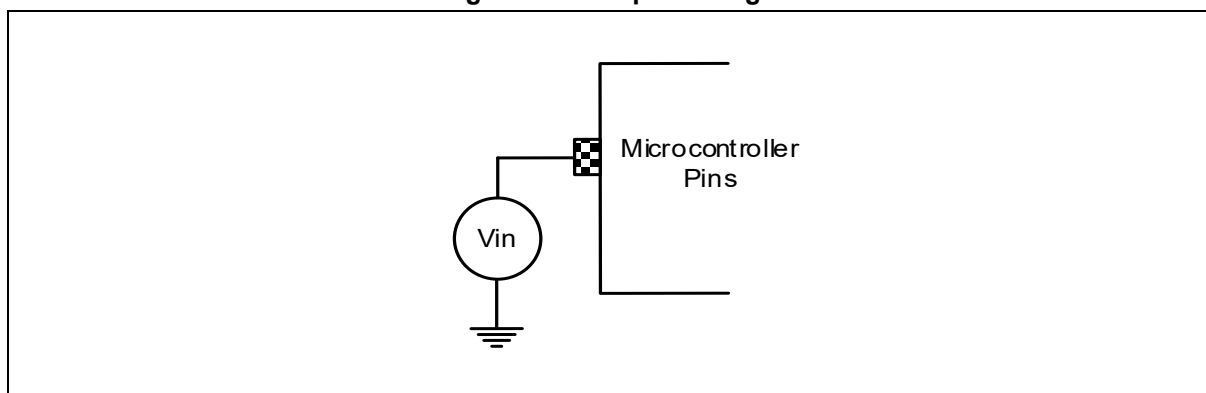
3.1.3 Typical Curves

Unless otherwise specified, all typical curves are provided only as design recommendations and are not tested.

3.1.4 Pin Input Voltage

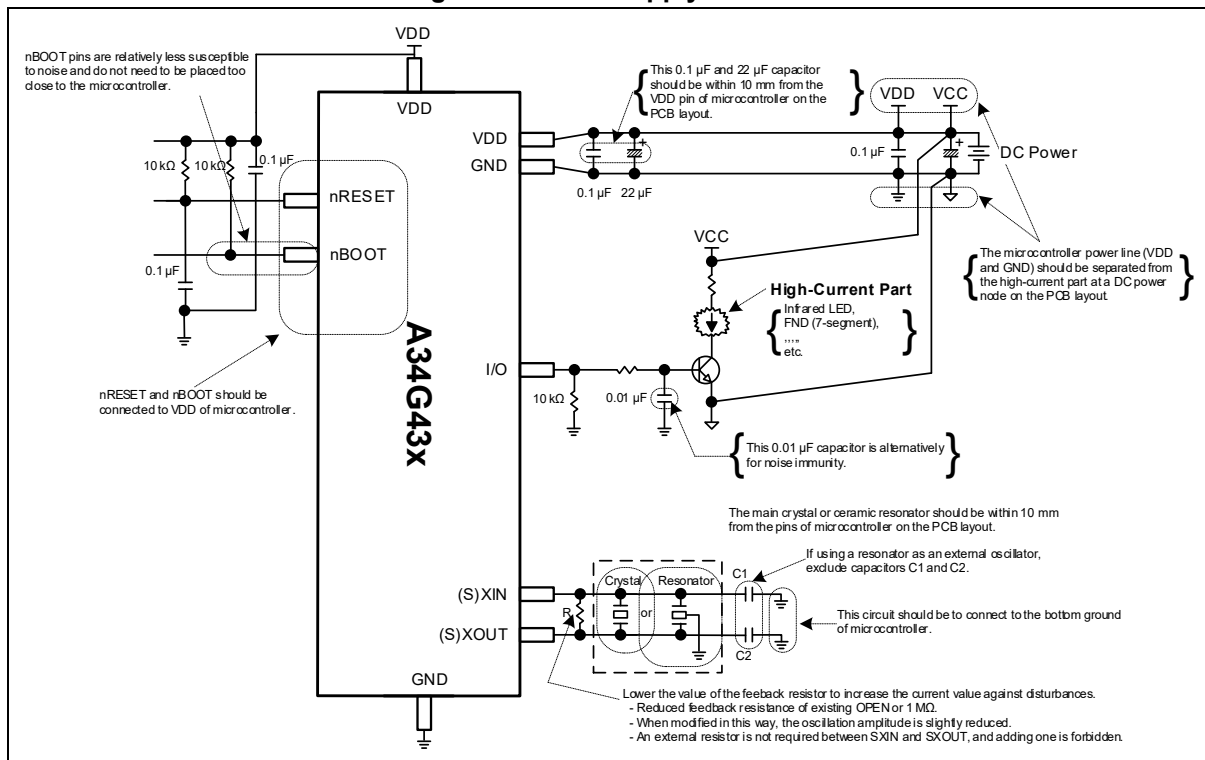
The input voltage measurement on a pin of the device is described in Figure 7.

Figure 7. Pin Input Voltage



3.1.5 Power Supply Scheme

Figure 8. Power Supply Scheme

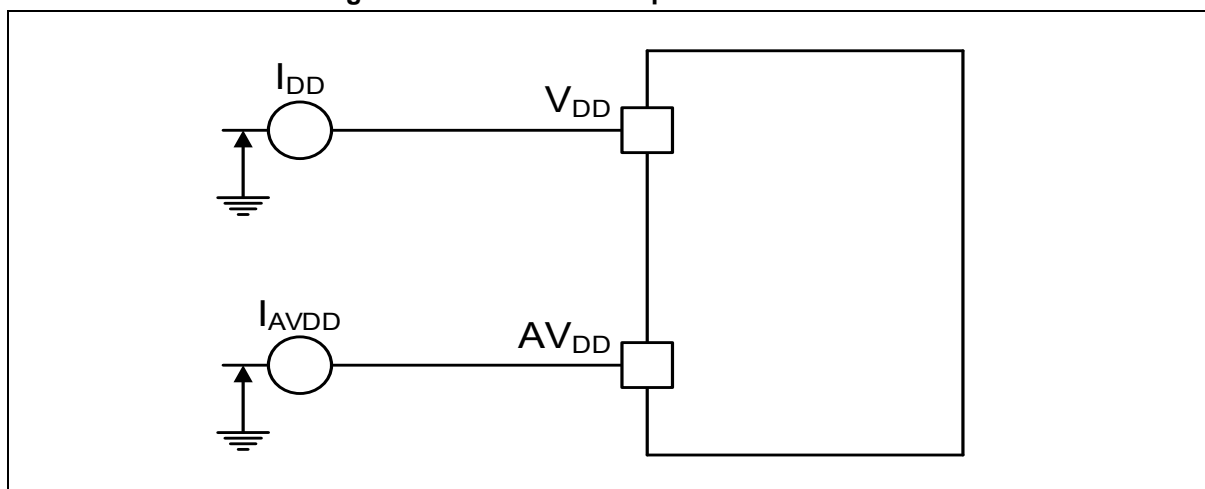


NOTE:

- Each power supply pair (VDD, GND, etc.) must be decoupled with filtering ceramic capacitors as shown above. These capacitors must be placed as close as possible to, or below, the appropriate pins on the underside of the PCB to ensure the good functionality of the device.

3.1.6 Current Consumption Measurement

Figure 9. Current Consumption Measurement



NOTE:

- The I_{DD} and I_{AVDD} parameters represent the total microcontroller consumption including the current supplying V_{DD} and AV_{DD} .

3.2 Absolute Maximum Ratings

Exceeding stresses specified in the Table 11 for voltage characteristics, Table 12 for current characteristics, or Table 13 for thermal characteristics may result in permanent damage to the device. The values listed in the tables are stress ratings only, and do not imply that the device will function properly under these conditions. Prolonged exposure to these maximum rating conditions may impact on the reliability of the device. It is important to operate the device within its specified maximum ratings to ensure reliable performance.

Table 11. Voltage Characteristics

Symbol	Ratings	Min.	Max.	Unit
VDD – GND ⁽¹⁾	External main supply voltage (including V _{DD} , AV _{DD})	–0.5	6.0	V
V _I ⁽²⁾	Input voltage on I/O	–0.5	Max (V _{DD} , AV _{DD}) + 0.5	V
V _O	Output voltage on I/O	–0.5	Max (V _{DD} , AV _{DD}) + 0.5	V

NOTES:

1. All main power (V_{DD}, AV_{DD}) and ground (GND, AGND) pins must always be connected to the external power supply, within the permitted range.
2. V_I maximum must always be respected.

Table 12. Current Characteristics

Symbol	Ratings	Max.	Unit
$\sum I_{VDD}$	Total current into sum of all V _{DD} power lines (source) ⁽¹⁾	–120	mA
$\sum I_{GND}$	Total current out of sum of all GND ground lines (sink) ⁽¹⁾	120	mA
I _{VDD(PIN)}	Maximum current into each V _{DD} power pin (source) ⁽¹⁾	–100	mA
I _{GND(PIN)}	Maximum current out of each GND ground pin (sink) ⁽¹⁾	100	mA
I _{IO(PIN)}	Output current sunk by any I/O	20	mA
	Output current sourced by any I/O and control pin	–10	mA
$\sum I_{IO(PIN)}$	Total output current sunk by sum of all I/Os and control pins ⁽²⁾	120	mA
	Total output current sourced by sum of all I/Os and control pins ⁽²⁾	–120	mA
I _{IL}	Input low leakage current	10	μA
I _{IH}	Input high leakage current	10	μA
P _T	Total power dissipation	600	mW

NOTES:

1. All main power (V_{DD}, AV_{DD}) and ground (GND, AGND) pins must always be connected to the external power supplies, within the permitted range.
2. This current consumption must be correctly distributed over all I/Os and control pins. The total output current must not be sunk/sourced between two consecutive power supply pins referring to high pin count LQFP packages.

Table 13. Thermal Characteristics

Symbol	Ratings	Value	Unit
T _{OP}	Operating temperature	-40 to +105	°C
T _{STG}	Storage temperature range	-55 to +125	°C
T _J	Maximum junction temperature	+105	°C

3.3 Operating Condition

3.3.1 General Operating Condition

Table 14. General Operating Condition

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{DD}	Supply voltage	-	2.7 ⁽¹⁾	-	5.5	V
AV _{DD}	Analog supply voltage	ADC, CMP, DAC used	2.7	-	5.5	V
f	Operating frequency	HSE	4	-	16	MHz
		LSE	-	32.768	-	kHz
		HSI (T _A =-40°C to 85°C)	15.52	16.00	16.48	MHz
		HSI (T _A =-40°C to 105°C)	15.20	16.00	16.80	MHz
		LSI	350	500	650	kHz
V _{IN}	I/O input voltage	5 V I/O	-0.3	-	V _{DD}	V
T _A	Ambient temperature	-	-40	-	105	°C
T _J	Junction temperature range	-	-	-	105	°C

NOTE:

1. When RESET is released, functionality is guaranteed down to V_{LVR0} Min.

3.3.2 Power-On Reset Characteristics

Table 15. POR Electrical Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
I _{DD}	Operating current	-	-	50	100	μA
V _{set}	POR set level	-	1.62	1.80	1.98	V
V _{reset}	POR reset level	-	1.44	1.60	1.76	V
t _R	V _{DD} voltage rising time	-	0.02	-	20	ms/V
t _F	V _{DD} voltage falling time	-	0.4	-	20	ms/V

NOTE:

1. Guaranteed by design.

3.3.3 Operating Condition at Power-Up/Power-Down

The parameters in Table 16 are derived from tests performed under the ambient temperature conditions summarized in Table 14.

Table 16. Operating Condition at Power-Up and Power-Down

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
t _{VDD}	V _{DD} rise time rate	-	0.02	-	20	ms/V
	V _{DD} fall time rate	-	0.4	-	20	ms/V

3.3.4 Embedded Reset and Power Control Block Characteristics (LVR, LVI)

The parameters in Table 17 and Table 18 are derived from tests performed under the ambient temperature conditions summarized in Table 14.

Table 17. Low-Voltage Reset Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{LVR0_F}	Detection level	Falling voltage	1.48	1.57	1.65	V
V _{LVR1_F}			1.58	1.66	1.75	
V _{LVR2_F}			1.69	1.78	1.87	
V _{LVR3_F}			1.78	1.88	1.98	
V _{LVR4_F}			1.88	1.99	2.09	
V _{LVR5_F}			2.00	2.11	2.21	
V _{LVR6_F}			2.16	2.28	2.39	
V _{LVR7_F}			2.31	2.43	2.56	
V _{LVR8_F}			2.52	2.66	2.80	
V _{LVR9_F}			2.85	3.00	3.16	
V _{LVR10_F}			2.97	3.14	3.30	
V _{LVR11_F}			3.35	3.53	3.71	
V _{LVR12_F}			3.52	3.71	3.90	
V _{LVR13_F}			3.82	4.03	4.23	
V _{LVR14_F}			3.99	4.21	4.42	
V _{LVR15_F}			4.24	4.47	4.70	
V _{LVR0_R}	Detection level	Rising voltage	1.57	1.66	1.74	V
V _{LVR1_R}			1.65	1.74	1.84	
V _{LVR2_R}			1.76	1.86	1.96	
V _{LVR3_R}			1.86	1.96	2.06	
V _{LVR4_R}			1.96	2.07	2.18	
V _{LVR5_R}			2.08	2.20	2.31	
V _{LVR6_R}			2.25	2.38	2.50	
V _{LVR7_R}			2.41	2.54	2.67	
V _{LVR8_R}			2.64	2.78	2.92	
V _{LVR9_R}			2.98	3.14	3.30	
V _{LVR10_R}			3.11	3.28	3.45	
V _{LVR11_R}			3.50	3.69	3.88	
V _{LVR12_R}			3.68	3.88	4.08	
V _{LVR13_R}			4.00	4.21	4.43	
V _{LVR14_R}			4.18	4.40	4.63	
V _{LVR15_R}			4.44	4.68	4.92	
-	Hysteresis	-	-	90	-	mV
-	Noise cancelling time	-	-	3.4	-	μs
I _{DD}	Operation current	-	-	20	35	μA
I _{DD, STOP}	Operation current (STOP)	-	-	10	1300	nA

NOTE:

1. Guaranteed by design.

Table 18. Low-Voltage Indicator Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{LV10_F}	Detection level	Falling voltage	1.48	1.57	1.65	V
V _{LV11_F}			1.58	1.66	1.75	
V _{LV12_F}			1.69	1.78	1.87	
V _{LV13_F}			1.78	1.88	1.98	
V _{LV14_F}			1.88	1.99	2.09	
V _{LV15_F}			2.00	2.11	2.21	
V _{LV16_F}			2.16	2.28	2.39	
V _{LV17_F}			2.31	2.43	2.56	
V _{LV18_F}			2.52	2.66	2.80	
V _{LV19_F}			2.85	3.00	3.16	
V _{LV110_F}			2.97	3.14	3.30	
V _{LV111_F}			3.35	3.53	3.71	
V _{LV112_F}			3.52	3.71	3.90	
V _{LV113_F}			3.82	4.03	4.23	
V _{LV114_F}			3.99	4.21	4.42	
V _{LV115_F}			4.24	4.47	4.70	
V _{LV10_R}	Detection level	Rising voltage	1.57	1.66	1.74	V
V _{LV11_R}			1.65	1.74	1.84	
V _{LV12_R}			1.76	1.86	1.96	
V _{LV13_R}			1.86	1.96	2.06	
V _{LV14_R}			1.96	2.07	2.18	
V _{LV15_R}			2.08	2.20	2.31	
V _{LV16_R}			2.25	2.38	2.50	
V _{LV17_R}			2.41	2.54	2.67	
V _{LV18_R}			2.64	2.78	2.92	
V _{LV19_R}			2.98	3.14	3.30	
V _{LV110_R}			3.11	3.28	3.45	
V _{LV111_R}			3.50	3.69	3.88	
V _{LV112_R}			3.68	3.88	4.08	
V _{LV113_R}			4.00	4.21	4.43	
V _{LV114_R}			4.18	4.40	4.63	
V _{LV115_R}			4.44	4.68	4.92	
-	Hysteresis	-	-	90	-	mV
-	Noise cancelling time	-	-	3.4	-	μs
I _{DD}	Operation current	-	-	20	35	μA
I _{DD, STOP}	Operation current (STOP)	-	-	10	1300	nA

NOTE:

1. Guaranteed by design.

3.3.5 Embedded Voltage Reference

Table 19. Embedded Internal Voltage Reference ⁽¹⁾

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{REFINT}	Internal reference voltage	-	0.58	0.60	0.62	V

NOTES:

1. Guaranteed by design.
2. The shortest sampling time is determined in the application by multiple iterations.

3.3.6 Supply Current Characteristics

The amount of current consumed by the device is determined by various factors and parameters, including but not limited to the operating voltage, ambient temperature, load on I/O pins, software configuration, operating frequency, switching rate of I/O pins, location of the program in memory, and the binary code being executed.

The current consumption is measured under the conditions specified in Table 20.

3.3.6.1 Supply Current Consumption

Table 20. Supply Current Characteristics

(Temperature: -40°C to +105°C)

Symbol	Operation Mode	Condition						Voltage Scaling	Typ.	Max.	Unit
		LSI (500 kHz)	HSI (16 MHz)	LSE (32.768 kHz)	HSE (8 MHz)	PLL (200 MHz)	f _{MCLK}				
I _{DD_RUN} (RUN)	Normal operation	STOP	RUN	STOP	STOP	STOP	f _{HSI}	5.0 V	11	-	mA
		STOP	STOP	STOP	RUN	STOP	f _{HSE}	5.0 V	11	-	
		STOP	STOP	STOP	RUN	RUN	f _{HSEPLL}	5.0 V	28	-	
		RUN	STOP	STOP	STOP	STOP	f _{LSI}	5.0 V	9	-	
		STOP	STOP	RUN	STOP	STOP	f _{LSE}	5.0 V	9	-	
I _{DD_SLEEP} (SLEEP)	SLEEP ⁽¹⁾	STOP	RUN	STOP	STOP	STOP	f _{HSI}	5.0 V	11	-	mA
		STOP	STOP	STOP	RUN	STOP	f _{HSE}	5.0 V	11	-	
		STOP	STOP	STOP	RUN	RUN	f _{HSEPLL}	5.0 V	20	-	
		RUN	STOP	STOP	STOP	STOP	f _{LSI}	5.0 V	9	-	
		STOP	STOP	RUN	STOP	STOP	f _{LSE}	5.0 V	9	-	
I _{DD_STOP2} (STOP 2)	DEEP-SLEEP ⁽¹⁾⁽²⁾	LVD (RUN)						5.0 V	6	-	mA
		LVD (STOP)						5.0 V	6	-	

NOTES:

1. All peripherals are turned off.
2. In DEEP-SLEEP mode, all clocks are disabled.

3.3.6.2 I/O System Current Consumption

The current consumption in the I/O system is due to the two components, Static and Dynamic.

3.3.6.3 I/O Static Current Consumption

All I/O pins used as inputs with pull-ups cause the current consumption when the pins remain Low from the outside. This current consumption value can be calculated simply using the pull-up/pull-down resistor value provided in the I/O port characteristics.

To estimate the current consumption at the output pins, users need to consider external Pull-downs or external load.

Additional I/O current consumption can be caused due to the I/Os configured as inputs if the medium voltage level is applied externally. This current is consumed by the input Schmitt-trigger circuit used to tell the input value. Unless the application requires a specific configuration, users can prevent this supply current consumption by configuring the I/Os in analog mode instead of using the Schmitt-trigger circuit. Specifically, input pins of the ADC and CMP must be configured as analog inputs.

3.3.6.4 I/O Dynamic Current Consumption

The I/Os used by an application also contribute to the current consumption. When an I/O pin switches, it uses the current from the I/O supply voltage to supply the I/O pin circuitry and to charge/discharge the capacitive load (internal or external) connected to the pin.

3.3.7 External Clock Source Characteristics

3.3.7.1 High-Speed External Clock Generated from Crystal/Ceramic Resonator

A high-speed external clock (HSE) can be supplied with crystal/ceramic resonator oscillators in the range from 4 to 16 MHz. All information provided in this section is based on the design simulation results obtained by the typical external components specified in Table 21.

To minimize the outputs in the application, the resonator and load capacitor must be placed as close to the oscillator pin as possible.

For more information on the distortion and stabilization time, and the resonator characteristics (frequency, package, and accuracy), contact the manufacturer of the crystal resonator.

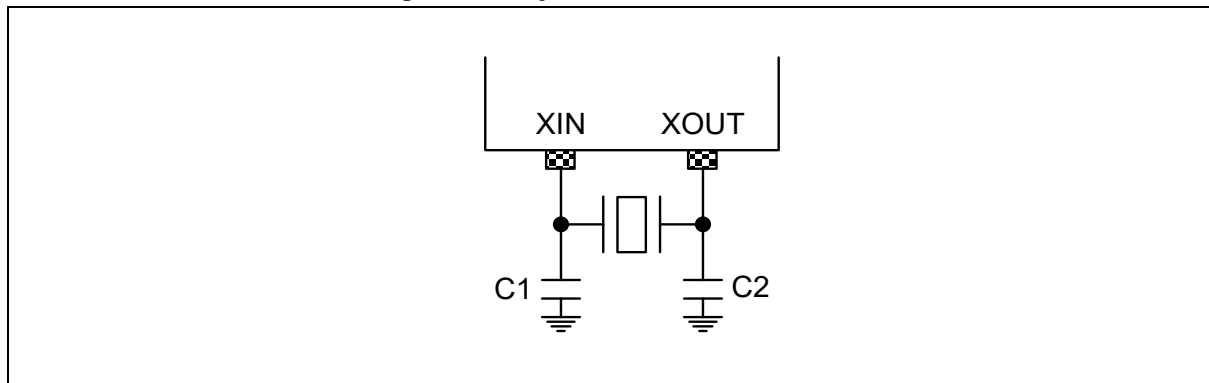
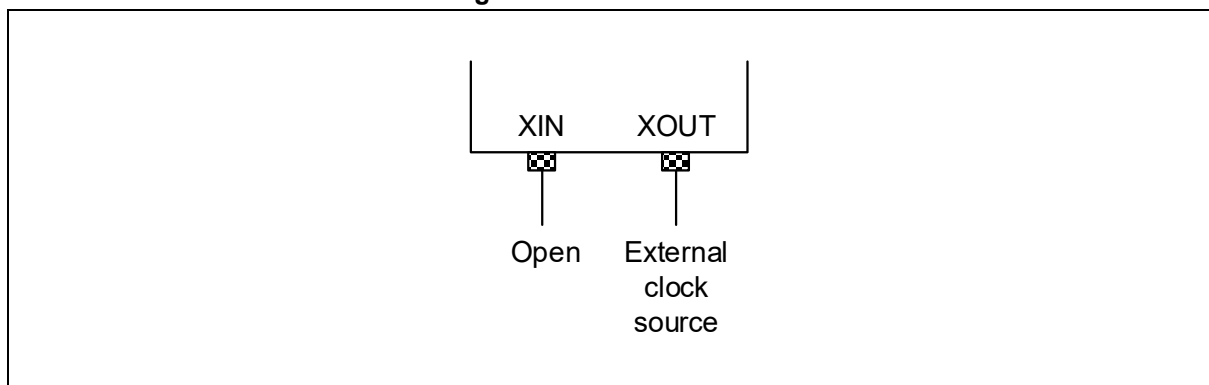
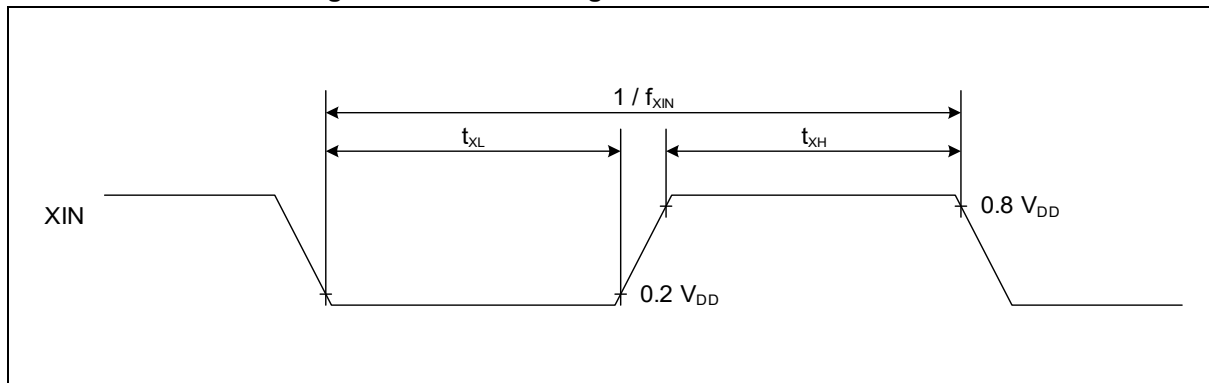
Table 21. HSE Oscillator Characteristics⁽¹⁾

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions ⁽²⁾	Min.	Typ.	Max.	Unit
V _{DD}	Operating voltage	-	2.7	5.0	5.5	V
I _{DD(HSE)}	HSE current consumption	-	-	-	2.5	mA
I _{STOP}	Power down current	-	-	9	930	nA
f _{OSC_IN}	Oscillator frequency	-	4	8	16	MHz
f _{OUT}	Output Frequency	V _{DD} ≥ 2.7 V ⁽⁴⁾⁽⁵⁾⁽⁶⁾⁽⁷⁾	4	-	16	MHz
V _{IL}	Crystal input (low)	-	-	-	0.2×V _{DD}	V
V _{IH}	Crystal input (high)	-	0.8×V _{DD}	-	-	V
V _{OL}	Crystal out (low)	-	-	-	0.2×V _{DD}	V
V _{OH}	Crystal out (high)	-	0.8×V _{DD}	-	-	V
C _L	Load Capacitor	f _{OUT} = 4 MHz	18	-	24	pF
		4 MHz < f _{OUT} ≤ 8 MHz	10	-	18	
		8 MHz < f _{OUT} ≤ 12 MHz	10	-	12	pF
		12 MHz < f _{OUT} ≤ 16 MHz	6	-	10	pF
R _F	Feedback resistor	V _{DD} = 5 V	0.7	1.0	1.3	MΩ
t _{SU(HSE)} ⁽³⁾	Startup time	-	-	1	-	ms

NOTES:

1. Guaranteed by design.
2. Resonator characteristics given by the crystal/ceramic resonator manufacturer.
3. t_{SU(HSE)} is the startup time measured from the moment it is enabled (by software) to a stabilized 8 MHz oscillation is reached. This value is measured for a standard crystal resonator, and it can vary significantly with the crystal manufacturer.
4. HSEISEL = 0x3, HSENFSEL = 0x0
5. HSEISEL = 0x2, HSENFSEL = 0x1
6. HSEISEL = 0x1, HSENFSEL = 0x2
7. HSEISEL = 0x0, HSENFSEL = 0x3

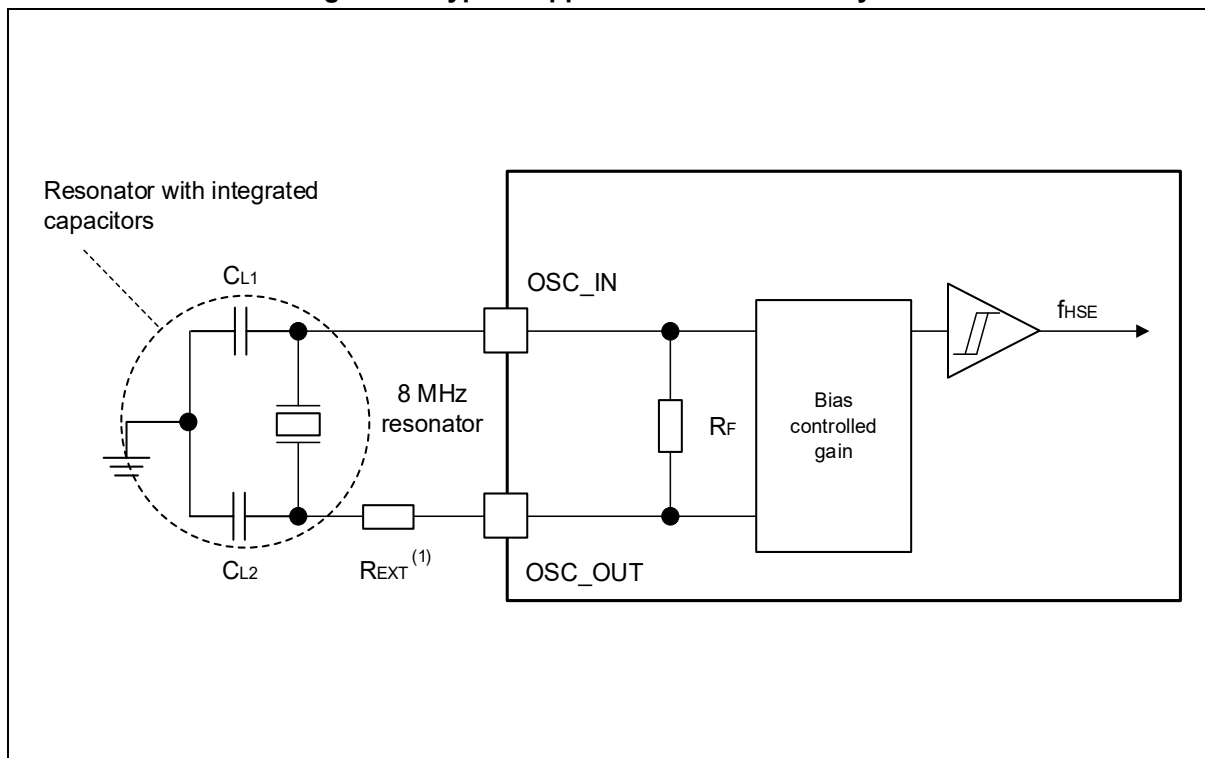
Figure 10. Crystal/Ceramic Oscillator**Figure 11. External Clock****Figure 12. Clock Timing Measurement at XIN Pin**

It is recommended to use high-quality external ceramic capacitors designed for high-frequency applications and selected to match the requirements of the crystal or resonator.

The C_{L1} and C_{L2} are usually the same size. The crystal manufacturers typically specify the load capacitance, a series combination of the C_{L1} and C_{L2} . When adjusting the size of the C_{L1} and C_{L2} , the PCB and microcontroller's pin capacitances must be considered together. (10 pF can be used as a rough estimate of the combined pins and board capacitance.)

Figure 13 shows a circuit diagram of a typical application with an 8 MHz crystal.

Figure 13. Typical Application with 8 MHz Crystal



3.3.7.2 Low-Speed External Clock Generated from Crystal Resonator

The low-speed external (LSE) clock can be generated using a crystal/ceramic resonator oscillator with 32.768 kHz. The information provided in this paragraph is based on characterization results obtained using typical external components listed in Table 22.

To minimize output distortion and startup stabilization time, it is recommended to place the resonator and load capacitors as close as possible to the oscillator pins in the application.

For further information on the resonator characteristics such as frequency, package, and accuracy, it is advisable to refer to the crystal resonator manufacturer.

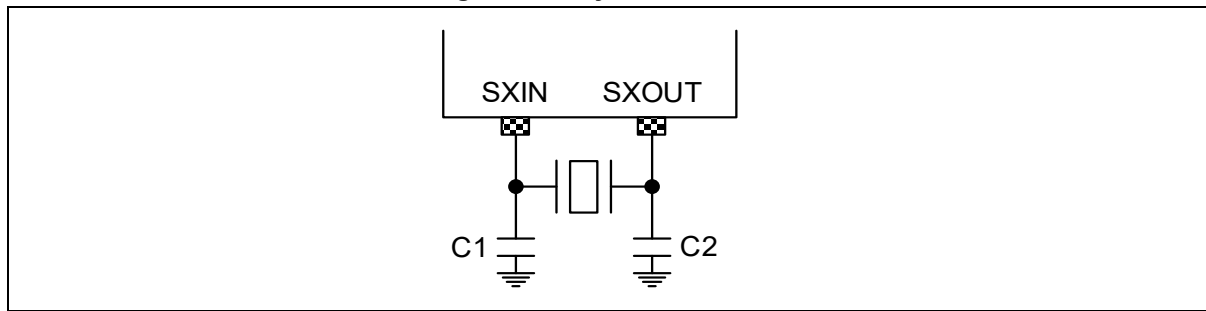
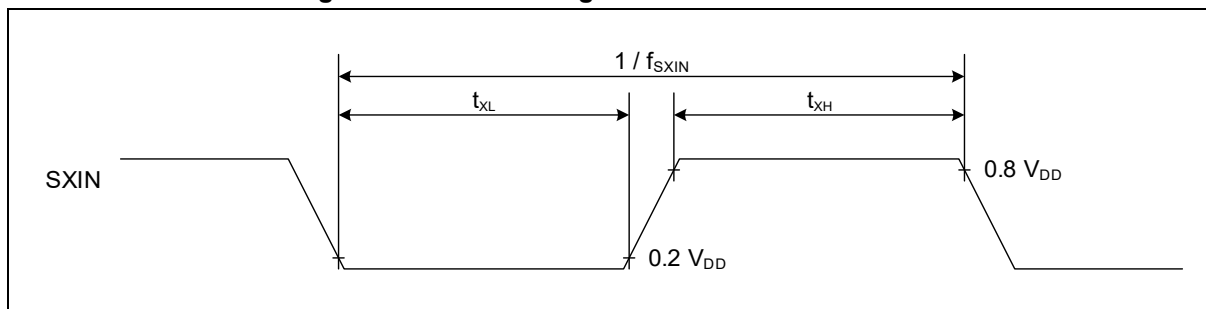
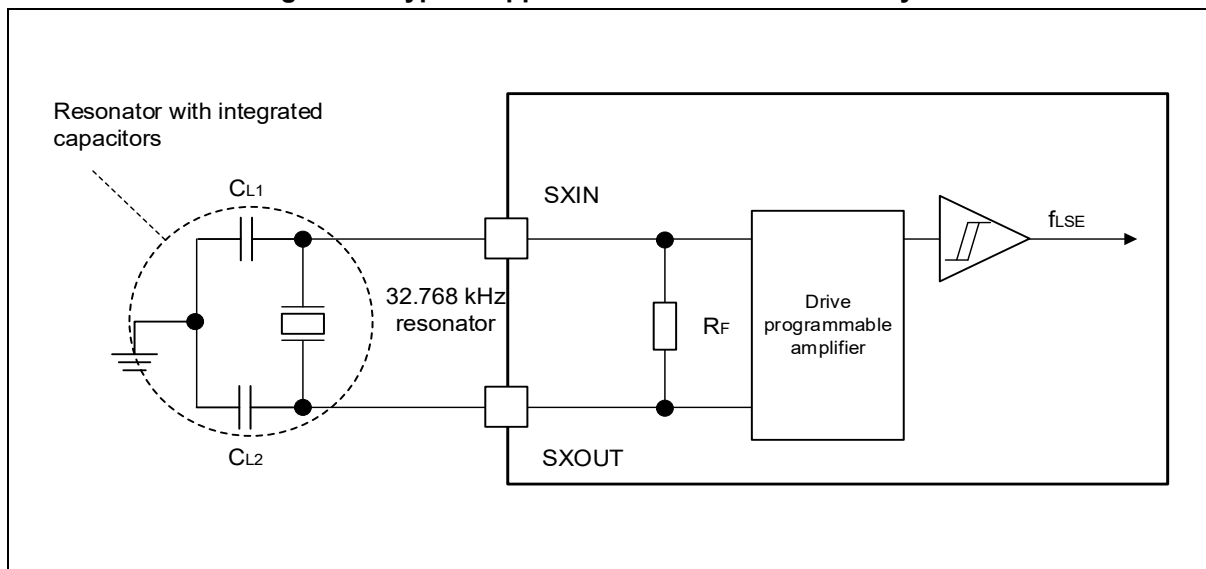
Table 22. LSE Oscillator Characteristics⁽¹⁾ ($f_{LSE} = 32.768$ kHz)

(Temperature: -40°C to $+105^{\circ}\text{C}$)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V_{DD}	Operating voltage	-	2.7	5.0	5.5	V
$I_{DD(LSE)}$	LSE current consumption	-	-	3	5	μA
I_{STOP}	Power down current	-	-	0.1	1	μA
f_{OSC_IN}	Oscillator frequency	-	-	32.768	-	kHz
V_{IL}	Crystal input (low)	-	-	-	$0.2 \times V_{DD}$	V
V_{IH}	Crystal input (High)	-	$0.8 \times V_{DD}$	-	-	V
V_{OL}	Crystal out (low)	-	-	-	$0.2 \times V_{DD}$	V
V_{OH}	Crystal out (high)	-	$0.8 \times V_{DD}$	-	-	V
C_L	Load Capacitor	-	5	-	15	pF
R_F	Feedback resistor	-	7	10	13	M Ω
$t_{SU(LSE)}^{(2)}$	Startup time	-	-	-	1	s

NOTES:

1. Guaranteed by design.
2. $t_{SU(LSE)}$ is the startup time measured from the moment it is enabled (by software) to a stabilized 32.768 kHz oscillation is reached. This value is measured for a standard crystal, and it can vary significantly with the crystal manufacturer.

Figure 14. Crystal Oscillator**Figure 15. Clock Timing Measurement at SXIN Pin****Figure 16. Typical Application With a 32.768 kHz Crystal****NOTE:**

1. An external resistor is not required between SXIN and SXOUT and it is forbidden to add one.

3.3.8 Internal Clock Source Characteristics

The parameters listed in Table 23 and Table 24 are obtained through tests conducted under ambient temperature and supply voltage conditions, which are specified in Table 14.

3.3.8.1 High-Speed Internal (HSI) RC Oscillator

Table 23. HSI Oscillator Characteristics⁽¹⁾

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V _{DD}	Operating voltage	-	2.7	-	5.5	V
f _{HSI}	HSI frequency	V _{DD} =5.0 V, T _A =30°C	15.808	16	16.192	MHz
Δ_F	Frequency error	@ 25°C	-1.2	-	1.2	%
		@ -40°C to +85°C	-3	-	3	%
		@ -40°C to +105°C	-5	-	5	%
DuCy (HSI16)	Duty cycle	-	45	-	55	%
Δ_{VDD} (HSI16)	HSI16oscillator Frequency drift over V _{DD}	V _{DD} =2.7 V to 5.5 V	-0.2	-	0.2	%
t _{SU} (HSI16)	HSI16oscillator start-up time	-	-	0.5	0.8	μs
t _{STAB} (HSI16)	HSI16oscillator stabilization time	-	-	1.8	2.1	μs
I _{DD} (HSI16)	HSI16oscillator power consumption	-	-	205	283	μA

NOTE:

1. Guaranteed by characterization results and design.

3.3.8.2 Low-Speed Internal (LSI) 500 kHz RC Oscillator

Table 24. LSI Oscillator Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions ⁽²⁾	Min.	Typ.	Max.	Unit
V _{DD}	Operating voltage		2.7	-	5.5	V
f _{LSI}	LSI frequency	V _{DD} = 2.7 to 5.5 V	350	500	650	kHz
t _{SU} (LSI) ⁽¹⁾	500 kHz oscillator start-up time (after LSI enabled)	V _{DD} = 5.0 V	-	5	10	μs
t _{STAB} (LSI) ⁽¹⁾	500 kHz oscillator stabilization time (after LSI enabled)	5% of target frequency	-	20	50	μs
I _{DD} (LSI) ⁽¹⁾	500 kHz oscillator power consumption	The LSI oscillator is enabled. V _{DD} =2.7 to 5.5 V	-	7.5	10	μA
		The LSI oscillator is disabled. V _{DD} =2.7 to 5.5 V	-	1	20	nA

NOTE:

1. Guaranteed by design.

3.3.9 PLL Characteristics

The parameters listed in Table 25 are obtained through tests conducted under ambient temperature and supply voltage conditions, which are specified in Table 14.

Table 25. PLL Characteristics⁽¹⁾

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
VDD	Operating voltage	-	2.7	-	5.5	V
I _{DD} (PLL)	PLL power consumption on V _{DD}	VCO freq = 400 MHz	-	1.5	-	mA
f _{PLLOUT}	PLL output clock	-	-	-	200	MHz
f _{VCO}	PLL VCO output	VCO frequency range	320	-	550	MHz
f _{DUTY}	Duty	-	40	-	60	%
f _{IN}	Input frequency	f _{PLLINCLK} / (PREDIV+1)	-	2	-	MHz
f _{PLLINCLK}	PLL block Input frequency	-	2	8	16	MHz
t _{JITTER}	P-P jitter	System clock 200 MHz	-	40	-	ps
t _{LOCK}	PLL lock time	-	-	100	150	μs

NOTE:

1. Guaranteed by design.
2. Take care of using the appropriate division factor (PREDIV) to obtain the specified PLL input clock value

3.3.10 Flash Memory Characteristics

Table 26. Code Flash Memory Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
t _{PROG}	Program time	-	-	-	30	μs
t _{PER}	Page erase time	-	-	-	4	ms
t _{SER}	Sector erase time	-	-	-	4	ms
t _{MER}	Mass (chip) erase time	-	-	-	8	ms
t _{READ}	Read access time	-	-	-	33	MHz
N _{FWE}	Endurance of write/erase	T _A =25°C, page unit	10,000	-	-	times
t _{RT}	Retention time	-	10	-	-	years

Table 27. Data Flash Memory Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
t _{PROG}	Program time	-	-	-	30	μs
t _{PER}	Page erase time	-	-	-	4	ms
t _{SER}	Sector erase time	-	-	-	4	ms
t _{MER}	Mass (chip) erase time	-	-	-	8	ms
t _{READ}	Read access time	N _{FWE} ≤ 10,000	-	-	33	MHz
		N _{FWE} ≤ 100,000			12.5	MHz
N _{FWE}	Endurance of write/erase	T _A =25°C, page unit	100,000	-	-	times
t _{RT}	Retention time	-	10	-	-	years

3.3.11 I/O Port Characteristics

Table 28. DC Electrical Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V_{IL}	Input low voltage	All GPIOs, nRESET, nBOOT	-	-	$0.3 \times V_{DD}$	V
V_{IH}	Input high voltage	All GPIOs, nRESET, nBOOT	$0.7 \times V_{DD}$	-	-	V
V_{OL}	Output low voltage	$V_{DD} = 3.3 \text{ V}$, $I_{OL1} = 1.5 \text{ mA}$, $I_{OL2} = 3 \text{ mA}^{(1)}$ $V_{DD} = 5 \text{ V}$, $I_{OL1} = 4 \text{ mA}$, $I_{OL2} = 8 \text{ mA}^{(1)}$	-	-	$0.2 \times V_{DD}$	V
V_{OH}	Output high voltage	$V_{DD} = 3.3 \text{ V}$, $I_{OH1} = -1.5 \text{ mA}$, $I_{OH2} = -3 \text{ mA}^{(1)}$ $V_{DD} = 5 \text{ V}$, $I_{OH1} = -4 \text{ mA}$, $I_{OH2} = -8 \text{ mA}^{(1)}$	$0.8 \times V_{DD}$	-	-	V
I_{OL1}	Output low current	$V_{DD} = 3.3 \text{ V}$, $V_{OL} = 0.2 V_{DD}$	-	-	$1.5^{(1)}$	mA
		$V_{DD} = 5 \text{ V}$, $V_{OL} = 0.2 V_{DD}$	-	-	$4^{(1)}$	mA
I_{OH1}	Output high current	$V_{DD} = 3.3 \text{ V}$, $V_{OH} = 0.8 V_{DD}$	$-1.5^{(1)}$	-	-	mA
		$V_{DD} = 5 \text{ V}$, $V_{OH} = 0.8 V_{DD}$	$-4^{(1)}$	-	-	mA
I_{OL2}	Output low current	$V_{DD} = 3.3 \text{ V}$, $V_{OL} = 0.2 V_{DD}$	-	-	$3^{(1)}$	mA
		$V_{DD} = 5 \text{ V}$, $V_{OL} = 0.2 V_{DD}$	-	-	$8^{(1)}$	mA
I_{OH2}	Output high current	$V_{DD} = 3.3 \text{ V}$, $V_{OH} = 0.8 V_{DD}$	$-3^{(1)}$	-	-	mA
		$V_{DD} = 5 \text{ V}$, $V_{OH} = 0.8 V_{DD}$	$-8^{(1)}$	-	-	mA
I_{IL}	Input low leakage current	All input ports	-10	-	-	μA
I_{IH}	Input high leakage current	All input ports	-	-	10	μA
R_{PU}	Pull-up resistor	All GPIOs, $V_{IN} = \text{GND}$	15	50	85	k Ω
R_{PD}	Pull-down register	All GPIOs, $V_{IN} = V_{DD}$	15	50	85	k Ω

NOTE:

- Each pin can adjust the drive strength, and the difference between $I_{OH/L1}$ and $I_{OH/L2}$ is due to the difference in strength.

3.3.11.1 General Input/Output Characteristics

Unless otherwise specified, the parameters in Table 29 are obtained through tests conducted under the conditions, which are specified in Table 14. All I/Os conform to CMOS standards.

Table 29. I/O Static Characteristics

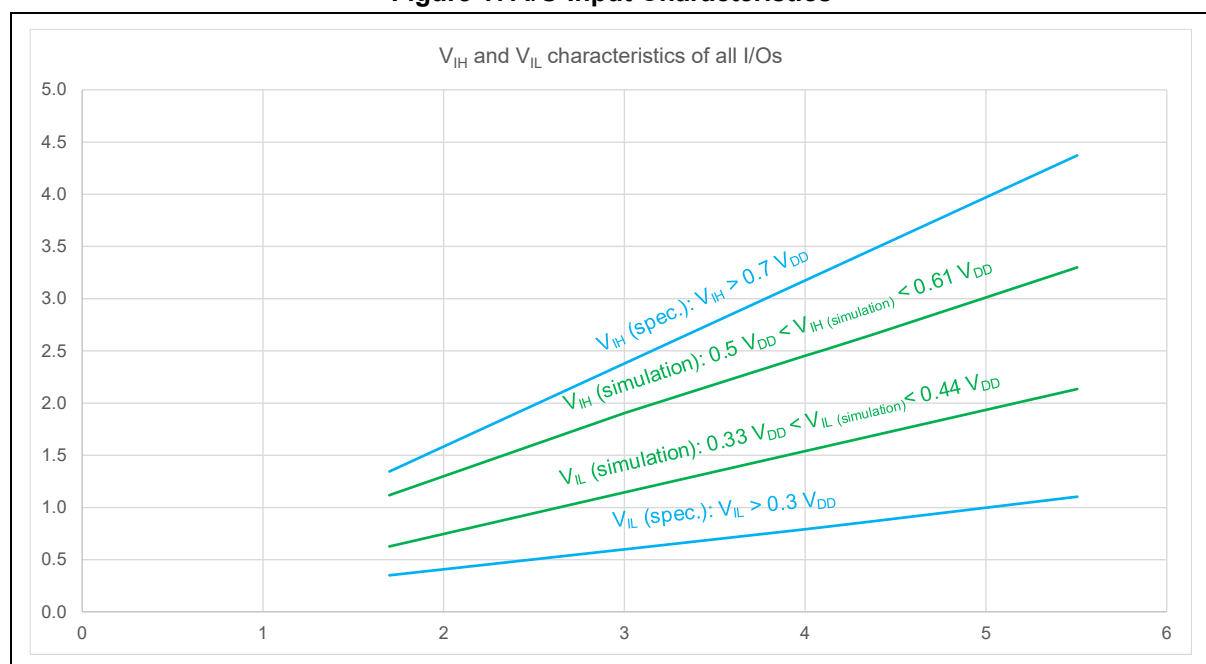
(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions		Min.	Typ.	Max.	Unit
$V_{IL}^{(1)(2)}$	I/O input low level voltage	5 V GPIO	$2.7\text{ V} \leq V_{DD} \leq 5.5\text{ V}$	-	-	$0.3 \times V_{DD}$	V
$V_{IH}^{(1)(2)}$	I/O input high level voltage	5 V GPIO	$2.7\text{ V} \leq V_{DD} \leq 5.5\text{ V}$	$0.7 \times V_{DD}$	-	-	V
$V_{HYS}^{(3)}$	Input hysteresis	5 V GPIO	$2.7\text{ V} \leq V_{DD} \leq 5.5\text{ V}$ $V_{DDTYPICAL} = 5\text{ V}$	-	$0.1 \times V_{DD}$	-	V
I_{leak}	Input leakage current ⁽³⁾	5 V GPIO	$0 \leq V_{IN} \leq V_{DD}$	-	-	± 10	μA
R_{PU}	Weak pullup equivalent resistor	$V_{IN} = \text{GND}$		15	50	85	k Ω
R_{PD}	Weak pulldown equivalent resistor	$V_{IN} = V_{DD}$		15	50	85	
C_{IO}	I/O pin capacitance	I/O pin capacitance		-	5	-	pF

NOTES:

1. Refer to Figure 17. I/O Input Characteristics
2. Data based on characterization results are not tested in production.
3. Guaranteed by design.

Figure 17. I/O Input Characteristics



NOTE:

1. X-axis is V_{DD}, Y-axis is V_{IH}/V_{IL} Spec.

3.3.11.2 Output Driving Current

The GPIOs are capable of sinking or sourcing currents up to ± 8 mA and can also sink or source currents up to ± 20 mA when the V_{OL}/V_{OH} limits are relaxed.

To ensure compliance with the absolute maximum ratings specified in chapter 3.2, it is necessary to limit the number of I/O pins driving current in the user application.

- The total current sourced by all I/O pins on VDD and the maximum operating current of the microcontroller on VDD (during Run mode) must not exceed the absolute maximum rating ΣI_{VDD} specified in Table 11.
- The total current sunk by all I/O pins on GND and the maximum operating current of the microcontroller on GND (during Run mode) must not exceed the absolute maximum rating ΣI_{GND} specified in Table 11.

3.3.11.3 Output Voltage Levels

Unless otherwise specified, parameters in Table 30 are obtained through tests conducted under ambient temperature and supply voltage conditions, which are specified in Table 14.

Table 30. Output Voltage Characteristics⁽¹⁾

(Temperature: -40°C to $+105^{\circ}\text{C}$)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit	
V _{OL}	Output low level voltage for an I/O pin	V _{DD} = 3.3 V, I _{OH1} = 1.5 mA, I _{OH2} = 3 mA	-	-	0.2×V _{DD}	V	
		V _{DD} = 5 V, I _{OH1} = 4 mA, I _{OH2} = 8 mA					
V _{OH}	Output high level voltage for an I/O pin	V _{DD} = 3.3 V, I _{OH1} = -1.5 mA, I _{OH2} = -3 mA	0.8×V _{DD}	-	-		
		V _{DD} = 5 V, I _{OH1} = -4 mA, I _{OH2} = -8 mA					
I _{OL1}	Output low current	V _{DD} = 3.3 V, V _{OL} = 0.2×V _{DD}	-	-	1.5		mA
		V _{DD} = 5 V, V _{OL} = 0.2×V _{DD}	-	-	4		mA
I _{OH1}	Output high current	V _{DD} = 3.3 V, V _{OH} = 0.8×V _{DD}	-1.5	-	-	mA	
		V _{DD} = 5 V, V _{OH} = 0.8×V _{DD}	-4	-	-	mA	
I _{OL2}	Output low current	V _{DD} = 3.3 V, V _{OL} = 0.2×V _{DD}	-	-	3	mA	
		V _{DD} = 5 V, V _{OL} = 0.2×V _{DD}	-	-	8	mA	
I _{OH2}	Output high current	V _{DD} = 3.3 V, V _{OH} = 0.8×V _{DD}	-3	-	-	mA	
		V _{DD} = 5 V, V _{OH} = 0.8×V _{DD}	-8	-	-	mA	

NOTE:

1. Guaranteed by design.

3.3.12 Digital-to-Analog Converter Characteristics

Table 31. DAC Electrical Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	unit
AV _{DD}	Analog Power Supply		2.7	-	5.5	V
R _L	Resistive load	Connected to GND	5	-	-	kΩ
C _L	Capacitive load	DAC output buffer on	-	-	50	pF
R _O	Output impedance	DAC output buffer off	23	30	50	kΩ
PSRR	Power supply rejection ratio	DAC output buffer on (CL=50 pF, RL=5 kΩ)	-	-60	-28	dB
I _{AVDD}	AVDD Current	DAC output buffer on (No load, middle code)	-	315	500	μA
		DAC output buffer off (No load, middle code)	-	-	0.2	μA
V _{DAC_OUT}	Conversion Voltage Range	DAC output buffer on	0.2	-	AV _{DD} -0.2	V
		DAC output buffer off	0	-	AVDD	V
T _{setting}	Setting time	DAC output buffer on (CL=50 pF, RL=5 kΩ)	-	-	3	μs
		DAC output buffer off (CL=50 pF, RL=5 kΩ)	-	-	3	μs
T _{wakeup}	Wakeup time from off state	DAC output buffer on (CL=50 pF, RL=5 kΩ)	-	4.2	7.5	μs
T _{W_to_W}	DAC output for a small variation of the input code (1-LSB)	-	1	-	-	μs
DNL	Differential nonlinearity	DAC output buffer on	-	-	±1.5	LSB
INL	Integrated nonlinearity	DAC output buffer on	-	-	±2	LSB
Offset	Offset error at full code (0x400)	DAC output buffer on	-	-	±1	LSB
		DAC output buffer off	-	-	±1	LSB
EGAIN	Gain error	DAC output buffer on	-	-	±0.2	%
		DAC output buffer off	-	-	±0.2	%
TUE	Total unadjusted error	DAC output buffer on	-	-	±5	LSB
		DAC output buffer off	-	-	±3	LSB
ENOB	Effective number of bits	DAC output buffer on	-	11	-	bits
		DAC output buffer off	-	11	-	bits

NOTE:

1. Guaranteed by design.

3.3.13 Comparator Characteristics

Table 32. Comparator Electrical Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	unit
V _{DD}	Operating voltage	-	2.7	-	5.5	V
I _{DD}	Operating current	-	-	100	450	μA
V _{IVR}	Input voltage range	-	0	-	5.5	V
V _{IO}	Input offset voltage	-	-5	-	+5	mV
t _D	Propagation delay	Overdrive voltage 100 mV	-	-	200	ns
V _{HYS}	Hysteresis	HYSSEL[1:0] = '00'	20	40	60	mV
		HYSSEL[1:0] = '01'	40	80	120	mV
		HYSSEL[1:0] = '10'	60	120	180	mV
		HYSSEL[1:0] = '11'	80	160	240	mV

NOTE:

1. Guaranteed by design.

3.3.14 Analog-to-Digital Converter Characteristics

Table 33. ADC Electrical Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	unit
AV _{DD}	Operating voltage	-	2.7	5	5.5	V
-	Resolution	-	-	-	12	bit
I _{DDA}	Operating current	AV _{DD} = 5.0 V @ f _{MCLK} = 25 MHz	-	1	-	mA
		AV _{DD} = 5.0 V @ f _{MCLK} = 42 MHz	-	1.5	-	mA
V _{AN}	Analog input range	-	GND	-	AV _{DD}	V
t _{CONV}	Conversion time	-	18 / f _{ACLK}	-	48 / f _{ACLK}	μs
f _{CONV}	Conversion frequency	@ AV _{DD} ≥ 3.7 V	-	-	1.5	MHz
		@ AV _{DD} ≥ 3.1 V	-	-	1.5	MHz
		@ AV _{DD} ≥ 2.7 V	-	-	1.0	MHz
f _{ACLK}	Operating frequency	-	-	-	42	MHz
INL	DC accuracy	-	-	±3	-	LSB
DNL		-	-	±3	-	LSB
ZOE	Zero offset error	-	-	±3	-	LSB
FSE	Full scale error	-	-	±3	-	LSB

NOTE:

1. Guaranteed by design.

3.3.14.1 General PCB Design Guidelines

The power supplier must be separated conforming to the scheme shown in Figure 8. The de-coupling capacitors across the V_{DD} and AV_{DD} are ceramic (good quality) capacitors and must be located as close as possible to the chip.

3.3.15 Communication Interfaces Characteristics

3.3.15.1 I2C Interface Characteristics

The I2C interface conforms to the timing requirements specified in the I2C-bus specification, in addition to the user timing requirements listed below:

- Standard mode (Sm): up to 100 kbps bit rate
- Fast mode (Fm): up to 400 kbps bit rate

The I2C timing requirements are guaranteed by design when the I2C peripheral is configured correctly and the I2C clock frequency is equal to or greater than the minimum value listed in the table below.

Table 34. I2C Characteristics

(Temperature: -40°C to +105°C)

Symbol	Parameter	Standard		Fast		Unit
		Min.	Max.	Min.	Max.	
f _{SCL}	Clock frequency	0	100	0	400	kHz
t _{SCLH}	Clock high pulse width	4.0	-	0.6	-	μs
t _{SCLL}	clock low pulse width	4.7	-	1.3	-	
t _{BF}	Bus free time	4.7	-	1.3	-	
t _{STSU}	Start condition setup time	4.7	-	0.6	-	
t _{STHD}	Start condition hold time	4.0	-	0.6	-	
t _{SPSU}	Stop condition setup time	4.0	-	0.6	-	
t _{SPHD}	Stop condition hold time	4.0	-	0.6	-	
t _{VD}	Output valid from clock	0	-	0	-	
t _{DIH}	Data input hold time	0	-	0	1.0	
t _{DIS}	Data input setup time	250	-	100	-	ns

3.3.15.2 SPI Interface Characteristics

Unless otherwise specified, the parameters in Table 35 are obtained through tests conducted under ambient temperature, f_{CLK} frequency and supply voltage conditions, which are specified in Table 14.

- Capacitive load $C = 30 \text{ pF}$
- Measurement points are performed at the CMOS level: $0.5 \times V_{DD}$

For more information about I/O alternate function characteristics (SS, SCK, MOSI, and MISO pins for SPI), refer to section 3.3.11.

Table 35. SPI Characteristics

(Temperature: -40°C to $+105^{\circ}\text{C}$)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
f_{SCK1}	SPI clock frequency with strength on	$2.7 \text{ V} \leq V_{DD} \leq 3.3 \text{ V}$	-	-	12	MHz
f_{SCK2}		$V_{DD} \leq 5.5 \text{ V}$	-	-	12	MHz
f_{SCK3}	SPI clock frequency with strength off	$2.7 \text{ V} \leq V_{DD} \leq 3.3 \text{ V}$	-	-	5	MHz
f_{SCK4}		$V_{DD} \leq 5.5 \text{ V}$	-	-	10	MHz
Duty	Duty cycle of SPI frequency (SCK)	Slave Mode	30	50	70	%

3.3.15.3 CANFD Interface

For more information about I/O alternate function characteristics (CANFD_TX and CANFD_RX pins for CANFD), refer to section 3.3.11.

3.3.15.4 QSPI Interface

Unless otherwise specified, the parameters in Table 36 and Table 37 are obtained through tests conducted under ambient temperature, f_{HCLK} frequency and supply voltage conditions, which are specified in Table 14.

- Capacitive load $C = 10$ or 30 pF
- Measurement points are performed at the CMOS level: $0.5 \times V_{DD}$

For more information about I/O alternate function characteristics (QSPI_IO0, QSPI_IO1, QSPI_IO2, QSPI_IO3, QSPI_SS and QSPI_SCK pins for QSPI), refer to section 3.3.11.

Table 36. QSPI Characteristics in SDR Mode⁽¹⁾

(Temperature: -40°C to $+105^{\circ}\text{C}$)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
f_{QCK} ($1/t_{QCK}$)	QSPI clock frequency with strength on	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ $CL=10\text{pF}$	-	-	40	MHz
		$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ $CL=30\text{pF}$	-	-	25	MHz
$t_{w(QCKH)}$	QSPI clock high and low time Even division	$PRD[7:0] = 1, 3, 5, 7, 9 \dots$	-	$t_{QCK} / 2$	-	ns
$t_{w(QCKL)}$		$* t_{QCK} = (PRD[7:0] + 1) / HCLK$	-	$t_{QCK} / 2$	-	ns
$t_{s(IN)}$	Data input setup time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	4	-	-	ns
$t_{h(IN)}$	Data input hold time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	2.5	-	-	ns
$t_{v(OUT)}$	Data output valid time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	2.5	-	-	ns
$t_{h(OUT)}$	Data output hold time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	0	-	-	ns

NOTE:

1. Guaranteed by design.

Figure 18. QSPI Timing Diagram in SDR Mode

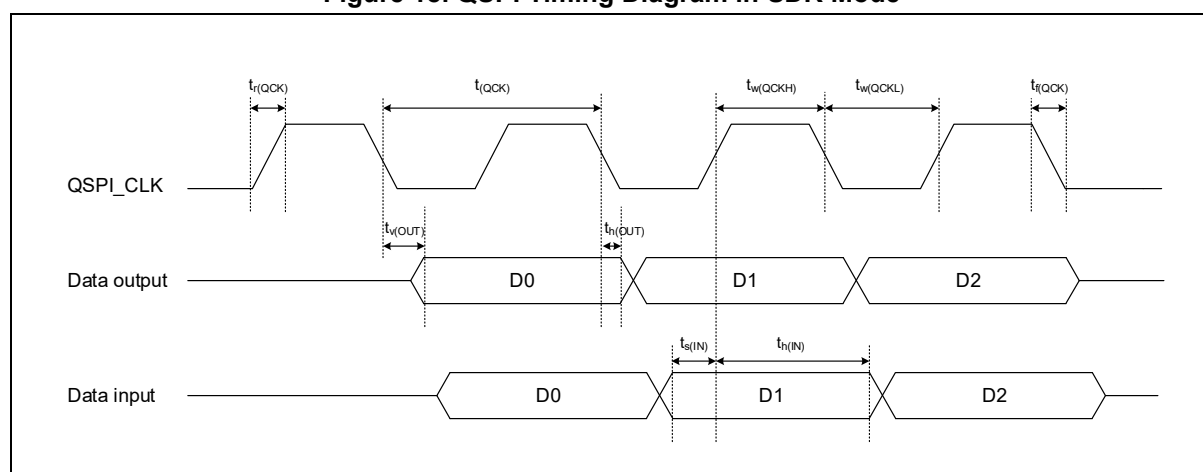


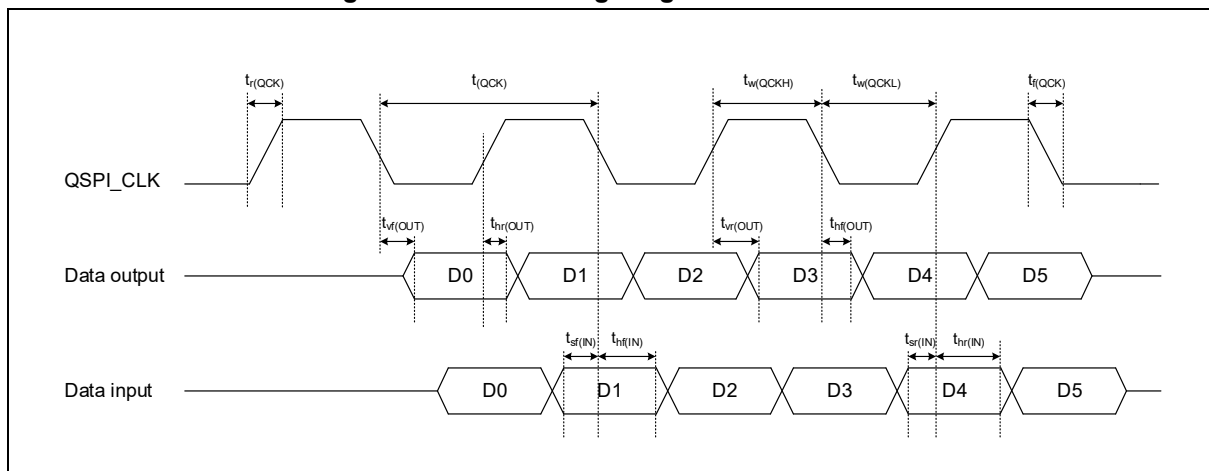
Table 37. QSPI Characteristics in DDR mode⁽¹⁾

(Temperature: -40°C to +105°C)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
f_{QCK} ($1/t_{QCK}$)	QSPI clock frequency with strength on	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ CL=10pF	-	-	33	MHz
		$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ CL=30pF	-	-	25	MHz
$t_{w(QCKH)}$	QSPI clock high and low time Even division	PRD[7:0] = 1, 3, 5, 7, 9... * $t_{QCK} = (\text{PRD}[7:0] + 1) / \text{HCLK}$	-	$t_{QCK} / 2$	-	ns
$t_{w(QCKL)}$			-	$t_{QCK} / 2$	-	ns
$t_{sr(IN)}, t_{sf(IN)}$	Data input setup time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	4	-	-	ns
$t_{hr(IN)}, t_{hf(IN)}$	Data input hold time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	2.5	-	-	ns
$t_{vr(OUT)}, t_{vf(OUT)}$	Data output valid time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	2.5	-	-	ns
$t_{hr(OUT)}, t_{hf(OUT)}$	Data output hold time	$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$	3	-	-	ns

NOTE:

1. Guaranteed by design.

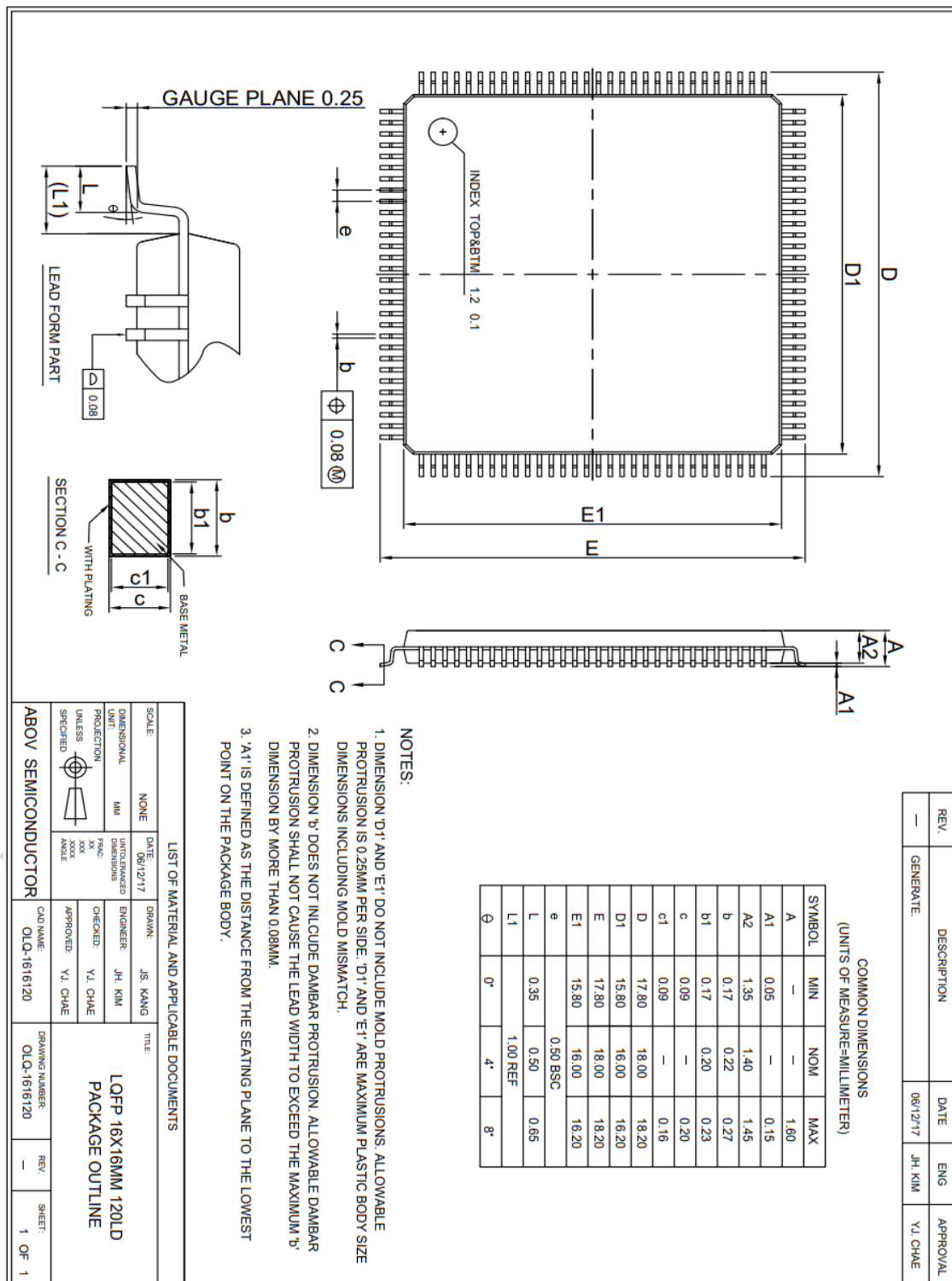
Figure 19. QSPI Timing Diagram in DDR mode

4. Package Information

4.1 120-LQFP Package Information

120-LQFP is a 120-pin, 16 x 16 mm quad flat package.

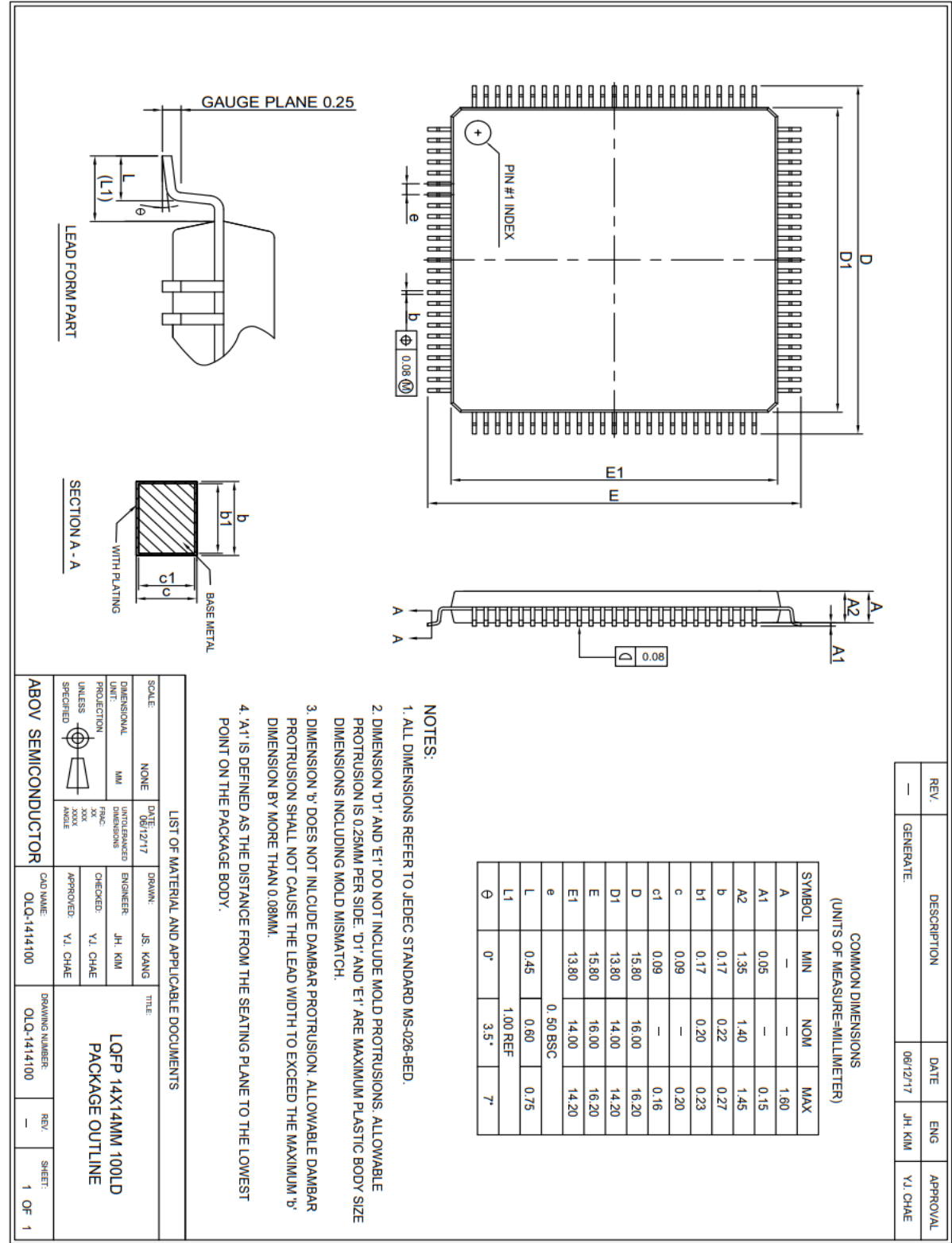
Figure 20. Package Dimension (120-LQFP)



4.2 100-LQFP Package Information

100-LQFP is a 100-pin, 14 x 14 mm quad flat package.

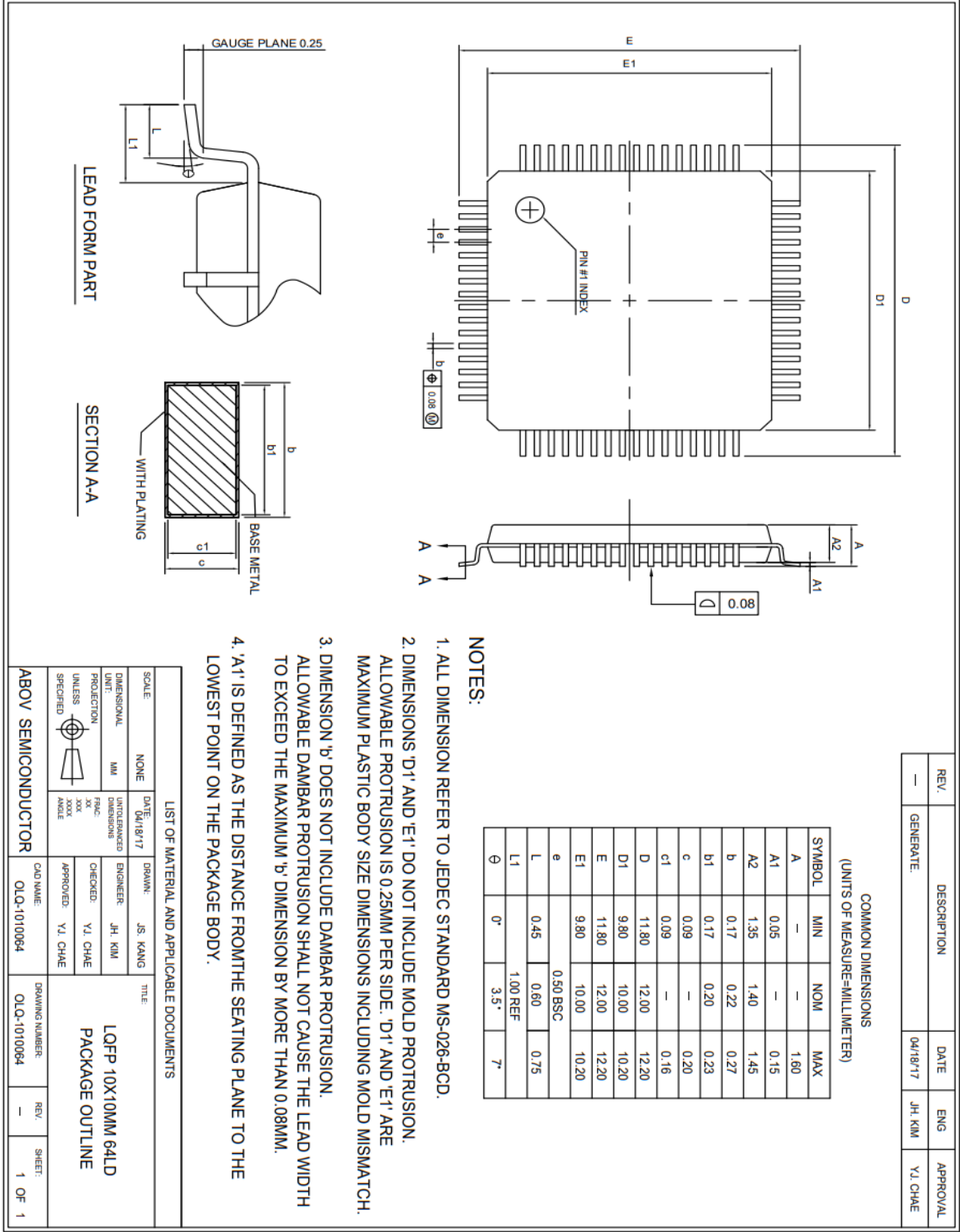
Figure 21. Package Dimension (100-LQFP)



4.3 64-LQFP Package Information

64-LQFP is a 64-pin, 10 x 10 mm quad flat package.

Figure 22. Package Dimension (64-LQFP)



5. Ordering Information

Figure 23. A34G43x Series Device Ordering Information

		A34G43	x	Y	L	2	N	(T)
Device Series Name								
A34G43	Arm Cortex-M4F based microcontroller							
Code Memory Size								
A	2048 KB							
0	1024 KB							
Pin Count								
Y	120-pin							
V	100-pin							
R	64-pin							
Package Type								
L	LQFP1, 0.50 mm pin pitch							
M	LQFP2, 0.65 mm pin pitch							
Temperature								
None	-40°C ~ +85°C							
2	-40°C ~ +105°C							
Internal Information								
N	Internal management code							
Packing								
None	Tray							
(T)	Tape & reel							

NOTE:

- For a list of available options (memory, package, and so on) or for further information on any aspect of this device, contact the nearest [ABOV sales office](#).

Revision History

Revision	Date	Notes
1.00	Dec. 02, 2025	Initial release

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